CLINICIAN ENDORSED CAREGIVER

A PROJECT REPORT

Submitted by

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In fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

INFORMATION TECHNOLOGY

SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY,

GANDHINAGAR Gujarat Technological University, Ahmedabad

Gujarat Technological University, Ahmedabad

April - 2026

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CERTIFICATE

This is to certify that the project report submitted along with the project entitled "**Clinician Endorsed Caregiver**" has been carried out by "**Sukhadiya Urvashi B**" under my guidance in fulfilment for the degree of Bachelor of Engineering in ***Department of Information Technology***, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2024-25.

Prof. Nirav Kamaliya	**Dr. Ramesh T. Prajapati**
Internal Guide	**Head of Department**
Internal Guide Sign:	**HOD Sign:**

Date: 31 March 2026

Experience Letter

To Whom So ever it may Concern

This is to certify that **Ms.Urvashi Sukhadiya** was associated / employed as Intern - PHP with our company: RainStream Technologies from 01/01/2026 to 31/03/2026.

During your tenure of approximately 03 months with our company, we found your conduct was good and approach to the work was found to be professional.

We wish you all the best in all your future endeavors!

For RainStream Technologies

Yours Sincerely,

Mehul Javia
CEO & Director
RainStream Technologies

DECLARATION

We hereby declare that the Internship report submitted along with the Internship entitled **Clinician Endorsed Caregiver** submitted in fulfilment for the degree of Bachelor of Engineering in **Information Technology** to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by us at ***Shree Swaminarayan Institute of Technology*** under the supervision of ***Prof. Nirav Kamaliya*** and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student: Sukhadiya Urvashi B.

Sign of Student :

ACKNOWLEDGEMENT

I wish to express my sincere gratitude to my **External guide** : **Priyanka Mattoo** for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank **Dr. Ramesh T. Prajapati (H.O.D. of IT Department)** for motivating me every time whenever I get confused, I would also like to thank my **Nirav Kamaliya** for helping me through my internship by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this Internship.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the company personals who gave me the permission to use and experience the valuable resources required for the Internship.

Thus, in conclusion to the above said, I once again thank the staff members of **Rainstream Technologies** for their valuable support in completion of the Internship.

Thank You

Name of Student
Urvashi Sukhadiya
Enrollment Number
231253116051

Date:

Signature of Student

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ABSTRACT

Caregivers play a crucial role in providing support to individuals with medical, psychological, or physical needs. However, they often face challenges in accessing reliable resources and guidance. This project **Clinician-Endorsed Caregiver,** aims to develop a web-based platform that provides verified, clinician-approved information for caregivers. The website will feature educational materials, self-care strategies, support groups, and interactive tools to assist caregivers in managing their responsibilities effectively. Over **11 million Americans** provide unpaid care for **6.7 million individuals** with dementia, yet most lack formal education or support. Clinicians struggle to meet the growing demand for caregiver education, and existing online resources are fragmented. To develop a **user-friendly, expert-approved** caregiver resource website that clinicians can easily recommend to family caregivers. Additionally, a **login system** will be integrated to personalize user experience and allow access to exclusive resources. By offering well-structured, evidence-based content, this project seeks to enhance the well-being of caregivers and improve patient care outcomes.

Keywords: Caregiver support, Clinician-endorsed resources, Healthcare information system, web development, MySQL database, integration Bootstrap, User authentication, Online resource platform

CHAPTER: 1

INTRODUCTION OF PROJECT & COMPANYPROFILE

1.1. Introduction

1.1.1. Company Profile

Rainstream Technologies is a technology-driven digital solutions company focused on delivering innovative, scalable, and result-oriented web and software services. The company specializes in helping businesses build a strong digital presence through modern design, development, and marketing strategies.

Rainstream Technologies serves startups, SMEs, and enterprises by combining creativity with technology to create impactful digital experiences. The company emphasizes quality, performance, and user-centric solutions to ensure business growth in a competitive digital landscape.

1.1.2. Company Products

- **Web Development:** Design and development of responsive, high-performance websites tailored to business needs using modern technologies.
- **Mobile App Development:** Creation of Android and iOS applications with intuitive UI/UX and scalable backend systems.
- **Custom Software Solutions:** Development of customized software systems to automate business processes and improve operational efficiency.
- **UI/UX Design:** Creative and user-focused interface designs that enhance user engagement and experience.
- **Digital Marketing Services:** SEO, social media marketing, paid advertising, and branding strategies to boost online visibility and lead generation.

- **E-commerce Development:** End-to-end e-commerce solutions including store setup, payment integration, and performance optimization.
- **Maintenance & Support:** Ongoing technical support, updates, and performance monitoring to ensure smooth functioning of digital products.

1.1.3. Company Mission and Vision

- **Company Mission**

- Promote a culture of innovation and creativity among students and faculty, inspiring them to think beyond conventional boundaries.
- Facilitate interdisciplinary collaboration to address real-world challenges and deliver effective, practical solutions.
- Provide state-of-the-art resources, mentorship, and infrastructure to support the entire lifecycle of innovative projects.
- Foster partnerships with industry, academia, and government organizations to promote knowledge sharing and technological growth.
- Empower students and innovators with the skills and confidence to become future leaders and change-makers in their respective fields.
- To serve as a location for the industrial collaborators to encourage their new Product Development in the campus using in-house facilities.
- To facilitate interdisciplinary design-focused education, research and entrepreneurial activities in order to create commercial opportunities and build partnerships between academics and industry.

- **Company Vision**

- To become a global leader in design innovation and technological advancement.
- To nurture creative minds and transform innovative ideas into impactful, real- world solutions.

- To establish a dynamic ecosystem where innovation thrives and contributes to sustainable and socially responsible advancements. GTU DIC envisions creating a dynamic ecosystem where innovation thrives, promoting sustainable and socially impactful technological advancements.
- To build a flagship program in the areas of design and innovation which can be replicated in other institutes/universities in our country.

1.2. Introduction of the Project

1.2.1. Purpose of the Project

The purpose of this project is to develop a clinician-endorsed, user-friendly online resource hub for caregivers. Many caregivers lack formal education or support, and clinicians struggle to provide consistent, high-quality information. This website will serve as a centralized platform where caregivers can access expert-reviewed educational materials, practical caregiving strategies, and self-care resources tailored to different stages of dementia.

- Provide reliable, high-quality care information approved by experts.
- Help clinicians easily recommend trusted resources to caregivers.
- Improve caregiver knowledge, confidence, and well-being through structured, accessible content.
- Establish a scalable model that can be adapted for other caregiver support needs.

1.2.2. Function Requirements

- User Registration & Login
- Role-Based Access
- Expert-Reviewed Content
- Search & Filter Options
- Self-Care & Well-being Section
- Surveys & Feedback Collection

1.2.3. Problems in Existing System

- **Lack of a Centralized Platform**
  - Caregivers must search multiple sources to find reliable information, leading to confusion and misinformation.
  - There is no single clinician-endorsed repository of dementia caregiving resources.
- **Inconsistent Quality & Reliability**
  - Many online resources lack expert approval, making it difficult to verify accuracy.
  - Outdated or misleading information can negatively impact caregiver decisions.
- **Limited Clinician Support & Accessibility**
  - Clinicians struggle to provide consistent education due to time constraints.
  - Many caregivers lack direct access to specialists or tailored support.
- **4. Poor User Experience & Navigation**
  - Existing websites may have complex layouts, making it hard for caregivers (especially elderly users) to find relevant information.
  - Resources are often not categorized based on dementia stages or caregiver needs.

1.2.4. Main Modules

- **User Management Module**
  - User Registration & Login
  - Role-Based Access
  - Profile Management
- **Resource Management Module**
  - Content Library:
  - Categorization System
  - Search & Filter Functionality

- **Caregiver Support Module**
  - Self-Care & Mental Health Tips:
  - Daily Care Strategies:
  - Frequently Asked Questions (FAQs):
- **Clinician Integration Module**
  - Resource Recommendation System:
  - Feedback & Reviews

CHAPTER: 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

2.1.1 Server-Side Requirements

- **Hardware Requirement**
  - I3 Processor or above.
  - 500 GB HDD or above
  - 8GB of RAM.
  - Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)
- **Software Requirement**
  - **Front End:** HTML, CSS, Bootstrap
  - **Back End:** PHP 7 and 8 and MySQL 5.0
  - **Server:** Apache Server 2.0
  - **Design Tool:** VS Code
  - **OS:** Window 7 or Any Compatible OS

2.1.2 Developer Side Requirements

- **Hardware Requirement**
  - I3 Processor or above.
  - 500 GB HDD or above
  - 8GB of RAM.
  - Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)
- **Software Requirement**
  - **Front End:** HTML, CSS, Bootstrap
  - **Back End:** PHP 7 and 8 and MySQL 5.0
  - **Server:** Apache Server 2.0
  - **Design Tool:** Sublime, VS Code
  - **OS:** Window 7 or Any Compatible OS

2.1.3 User Side Requirements

- **Hardware Requirement**
  - I3 Processor or above.
  - 500 GB HDD or above
  - 8GB of RAM.

- Keyboard (Normal or Multimedia) and Mouse (Compatible Mouse)

- **Software Requirement**
  - **Front End:** HTML, CSS, Bootstrap
  - **Back End:** PHP 7 and 8 and MySQL 5.0
  - **Server:** Apache Server 2.0
  - **Design Tool:** Sublime, VS Code
  - **OS:** Window 7 or Any Compatible

CHAPTER: 3

WORKSHEET REPORT

3.1 Work Sheet Report:

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	o **Front End:** HTML, CSS, Bootstrap o **Back End:** PHP and MySQL o **Server:** Apache Server 2.0 o **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
1	29/12/2025 To 02/01/2026	Researched and finalized the project topic. Explored similar caregiver resource websites for inspiration.	Completed	
2	05/01/2026 To 9/01/2026	Planned website structure and created a wireframe. Started homepage development using HTML & CSS.	Completed	

SUGGESTED 15 DAYS WORK SHEET REPORT

Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	o **Front End:** HTML, CSS, Bootstrap o **Back End:** PHP 7 and MySQL o **Server:** Apache Server 2.0 o **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
3	12/01/2026 To 16/01/2026	Built About & Contact pages. Designed navigation bar & footer using Bootstrap.	Completed	
4	19/01/2026 To 23/01/2026	Created forms for caregiver login/signup. Styled forms using Bootstrap.	Completed	

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	○ **Front End:** HTML, CSS, Bootstrap ○ **Back End:** PHP 7 and MySQL ○ **Server:** Apache Server 2.0 ○ **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
5	26/01/2026 To 30/01/2026	Set up MySQL Workbench. Designed database schema for login form.	Completed	
6	02/02/2026 To 06/02/2026	Connected MySQL Workbench with basic queries. Reviewed and refined UI.	Completed	

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	○ **Front End:** HTML, CSS, Bootstrap ○ **Back End:** PHP 7 and MySQL ○ **Server:** Apache Server 2.0 ○ **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
7	09/02/2026 To 13/02/2026	Create Models for Subscribers , Caregiver and Relevant entities	Completed	
8	16/02/2026 To 20/02/2026	Develop homepage Using Bootstrap (Navigation , Hero section) Design Login Form	Completed	

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	○ **Front End:** HTML, CSS, Bootstrap ○ **Back End:** PHP 7 and MySQL ○ **Server:** Apache Server 2.0 ○ **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
9	23/02/2026 To 27/02/2026	Build Multiple Pages for different section of the website	Completed	
10	02/03/2026 To 06/03/2026	Front-end Development layout & Design	Completed	

SUGGESTED 15 DAYS WORK SHEET REPORT	
Student Name:	**Sukhadiya Urvashi**
Enrollment No:	**231253116051**
Internship/Project Title	**Clinician-Endorsed Caregiver**
Tools and Technologies	○ **Front End:** HTML, CSS, Bootstrap ○ **Back End:** PHP 7 and MySQL ○ **Server:** Apache Server 2.0 ○ **Design Tool:** VS Code
Company/ Organization Name	**GTU-Design Innovation Center (DIC)**

Student's Activity Details:

Week Number	Start Date to End Date	Tasks to be assigned	Tasks to be completed	Remarks
11	09/03/2026 To 13/03/2026	Develop a Page new guidelines Posts	Completed	
12	16/03/2026 To 31/03/2026	Add Email Functionality For Contact & Reply Design. And testing	Completed	

CHAPTER: 4

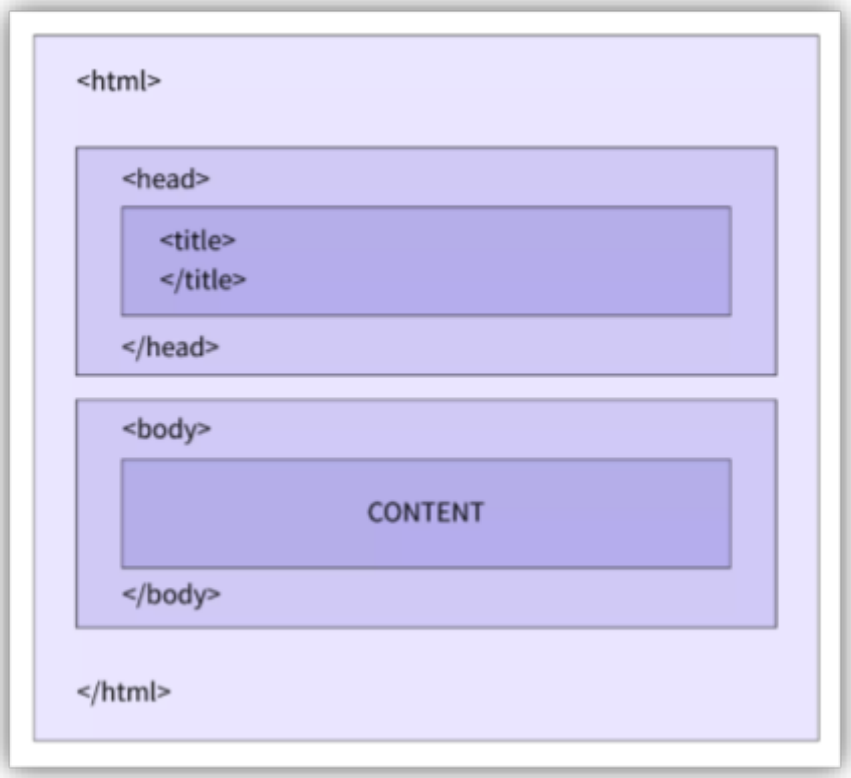
FRONT END OF SYSTEM

4.1 About Front End

- Front end development refers to the part of web development that focuses on creating the user interface and experience of a website or web application.
- It involves implementing designs provided by UI/UX designers using languages such as HTML, CSS, and JavaScript.

4.1.1 about HTML

- HTML stands for Hyper Text Markup Language.
- HTML describes the structure of a Web page.
- HTML consists of a series of elements.
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- **HTML Page Structure**



- Any HTML document mainly comprises two parts, the head and the body.
- The Head includes the additional information of the document, apart from the one displayed on the web page.
- It includes the HTML version, the title, and also the metadata.
- And the information that will be displayed on the web page is put inside the Body tag.
- HTML, or Hypertext Markup Language, is the standard language used to create and design web pages.
- It provides the structure and content of a webpage by using a system of markup tags and attributes.

- **Here's a breakdown of HTML's key components and concepts:**
  - **Elements:**
    - HTML documents are composed of elements, which are represented by tags enclosed in angle brackets (< and >).
    - Elements can have attributes that provide additional information about them.
  - **Tags**:
    - Tags are used to define the beginning and end of an element.
    - Some elements are self-closing, meaning they don't have content and close themselves within the opening tag.
  - **Attributes:**
    - Attributes provide additional information about an element and are specified within the opening tag.
    - They are comprised of a name and a value, separated by an equals sign (=).
  - **Document Structure:**
    - HTML documents have a specific structure defined by a hierarchy of elements.

- The basic structure typically includes a <!DOCTYPE> declaration,
  <html>, <head>, and <body> elements.

- **Semantic Markup:**
  - HTML provides semantic elements that convey the meaning and structure of content to both browsers and developers
  - These elements include headings (<h1> to <h6>), paragraphs (<p>), lists (<ul>, <ol>, <li>), and more.
- **Links and Image:**
  - HTML allows you to include hyperlinks (<a>) to navigate between pages or sections within a page, and images (<img>) to display graphics on a webpage.

4.1.2 about CSS

- CSS stands for Cascading Style Sheets.
- CSS describes how HTML elements are to be displayed on screen, paper, or in other media
- CSS saves a lot of work. It can control the layout of multiple web pages all at once
- External stylesheets are stored in CSS files
- CSS, or Cascading Style Sheets, is a style sheet language used to describe the presentation of a document written in HTML or XML (including XML dialects like SVG or XHTML). CSS describes how elements should be rendered on screen, in print, or even spoken aloud.

- **Here's a deeper look at CSS:**
  - **Selectors:**
    - CSS selectors are patterns used to select the elements you want to style.
    - They can target HTML elements, classes, IDs, attributes, and more.

- **Properties and Values:**
  - CSS properties are the stylistic attributes you can apply to selected elements, and values determine how those properties are applied.
- **Cascade and Specificity:**
  - The "C" in CSS stands for "Cascading," meaning that styles can cascade from parent elements to child elements. CSS also follows rules of specificity.
- **Units and Values:**
  - CSS supports various units for specifying sizes, including pixels (px), percentages (%), ems (em), and more.
  - It also supports different value types such as colors, lengths, and keywords.
- **Box Model:**
  - CSS treats each HTML element as a box, comprising content, padding, border, and margin.
  - Understanding the box model is crucial for layout and positioning of elements on a webpage.
- **Responsive Design:**
  - CSS allows developers to create responsive designs that adapt to different screen sizes and devices using techniques like media queries, flexbox, and CSS grid.
- **Preprocessors and Frameworks:**
  - CSS preprocessors like Sass and Less extend the capabilities of CSS by adding features like variables, mixing, and nested rules.
  - CSS frameworks like Bootstrap and Foundation provide pre- written CSS and HTML templates to streamline

CSS plays a fundamental role in web development by controlling the look and feel of web pages, making them visually appealing and user-friendly. It works hand- in-hand with HTML and often JavaScript to create dynamic and interactive web experiences

CHAPTER: 5

BACK END OF SYSTEM

5.1 About PHP

PHP (Hypertext Preprocessor) is a widely-used open-source scripting language that is particularly suited for web development and can be embedded into HTML. It was originally created by Danish-Canadian programmer Rasmus Lerdorf in 1994 and has since evolved into a powerful tool for creating dynamic web pages and web applications.

- **Server-Side Scripting:** PHP is primarily a server-side scripting language, meaning it runs on a web server. When a user requests a PHP page, the PHP code is executed on the server, and the resulting HTML is sent to the client's web browser.
- **Embeddable:** PHP code can be embedded directly into HTML, allowing for dynamic content generation. This makes it easy to create web pages that change content based on user input or other factors.
- **Open Source:** PHP is open-source software, which means it is freely available for anyone to use and modify. This has led to a large and active community of developers who contribute to its development and create a vast ecosystem of libraries, frameworks, and tools.
- **Cross-Platform Compatibility:** PHP is compatible with all major operating systems, including Windows, macOS, Linux, and Unix variants. It also supports a wide range of web servers, including Apache, Nginx, and Microsoft IIS.
- **Database Integration:** PHP has built-in support for working with databases, particularly MySQL, which makes it well-suited for building database-driven web applications. However, it also supports other databases such as PostgreSQL, SQLite, and MongoDB.
- **Frameworks and CMS:** There are several popular PHP frameworks and content management systems (CMS) that simplify the process of building web applications and websites. Examples include Laravel, Symfony, CodeIgniter, and WordPress.
- **Security:** Like any programming language, PHP is susceptible to security vulnerabilities if not used properly. However, PHP provides features and functions to help developers write secure code, such as input validation and output escaping functions.
- **Performance:** PHP's performance has improved significantly over the years, especially

with the introduction of PHP 7 and later versions, which include performance enhancements and optimizations.

Overall, PHP remains a popular choice for web development due to its ease of use, flexibility, and extensive ecosystem of tools and resources.

5.2 Why Use PHP

There are several reasons why PHP continues to be a popular choice for web development:

- **Ease of Learning and Use:** PHP has a relatively simple and straightforward syntax that is easy for beginners to grasp. Its syntax is similar to C and Perl, which many developers are already familiar with. This simplicity makes it accessible to developers of varying skill levels.
- **Wide Adoption:** PHP powers a significant portion of the web, with millions of websites and web applications built using PHP. This widespread adoption means there is a large community of developers who can provide support, share knowledge, and contribute to the development of tools and libraries.
- **Flexibility:** PHP is highly flexible and can be used to build a wide range of web applications, from small personal websites to large-scale enterprise systems. It supports various programming paradigms, including procedural programming, object-oriented programming (OOP), and functional programming.
- **Integration with Web Servers and Databases:** PHP works seamlessly with popular web servers like Apache and Nginx, as well as a variety of databases such as MySQL, PostgreSQL, SQLite, and MongoDB. This makes it easy to build database-driven web applications with PHP.
- **Rich Ecosystem**: PHP has a vast ecosystem of frameworks, libraries, and tools that streamline the development process and provide solutions to common problems. Frameworks like Laravel, Symfony, and CodeIgniter offer robust features for building web applications, while CMS platforms like WordPress and Joomla simplify the development of content-based websites.
- **Rapid Development:** PHP's ability to embed code directly into HTML allows for rapid development of dynamic web pages. Developers can mix PHP code with HTML markup to create templates that generate dynamic content based on user input or other factors
- **Cost-Effectiveness:** PHP is open-source software, which means it is free to use and distribute. There are no licensing fees associated with PHP, making it a

cost-effective choice for businesses and developers.

- **Performance Improvements:** With each new release, PHP continues to improve its performance and efficiency. PHP 7 introduced significant performance enhancements, including improved memory usage and faster execution speeds. Subsequent versions have built upon these improvements, making PHP a competitive option for high- performance web applications.

Overall, PHP offers a compelling combination of simplicity, flexibility, performance, and community support, making it an excellent choice for web development projects of all sizes.

5.3 About Back End

The backend, also known as server-side, refers to the part of a web application or software that operates behind the scenes and is responsible for processing requests, managing data, and generating responses to be sent to the client-side (frontend). Here are some key aspects of backend development:

- **Data Management:** One of the primary functions of the backend is managing data. This includes tasks such as storing, retrieving, updating, and deleting data from databases. Backend developers work with databases such as MySQL, PostgreSQL, MongoDB, or others to ensure efficient data storage and retrieval.

- **Business Logic:** The backend contains the business logic of the application, which defines how data is processed and manipulated based on the application's rules and requirements. This may involve implementing algorithms, calculations, validations, and other operations necessary to achieve the desired functionality.

- **Server-Side Scripting**: Backend development often involves writing server-side code using programming languages like PHP, Python, Ruby, Java, Node.js, or others. This code runs on the server and handles client requests, executes business logic, interacts with databases, and generates dynamic content
- **APIs (Application Programming Interfaces):** Backend developers design and implement APIs that allow different components of a system to communicate with eachother. APIs define the protocols and rules for interacting with the backend, enabling frontend applications, mobile apps, and other services to access and manipulate data.
- **Security:** Backend developers are responsible for implementing security measures to protect the application and its data from unauthorized access, data

breaches, and other security threats. This includes techniques such as authentication, authorization, encryption, input validation, and secure coding practices.

- **Scalability and Performance:** Backend developers design and optimize the backend architecture to ensure scalability and performance, especially for applications with high traffic or large datasets. This may involve using caching mechanisms, load balancing, horizontal scaling, and other techniques to handle increased demand and maintain responsiveness.
- **Integration:** Backend systems often need to integrate with external services, third- party APIs, payment gateways, and other systems. Backend developers handle the integration process, ensuring seamless communication and interoperability between different components of the application ecosystem.
- **Monitoring and Maintenance:** Backend developers monitor the health and performance of the backend systems, identify and troubleshoot issues, and perform routine maintenance tasks such as database backups, software updates, and security patches.

In summary, backend development is crucial for building robust, scalable, and secure web applications and software systems. It involves handling data management, implementing business logic, writing server-side code, designing APIs, ensuring security, optimizing performance, integrating with external services, and maintaining the health of backend systems.

5.3.1 about MySQL

MySQL is an open-source relational database management system (RDBMS) that is widely used for web development and other applications requiring a robust, scalable, and reliable database solution. It's known for its speed, ease of use, and comprehensive support for various programming languages

- **Here are some key aspects of MySQL:**
  - **Relational Database Management System:** MySQL follows the relational database model, organizing data into tables with rows and columns. It supports SQL for querying and managing the database.
  - **Open Source:** MySQL is an open-source software, meaning it is freely available for anyone to use, modify, and distribute. This has contributed to its widespread adoption and vibrant community of developers and contributors.

- **Cross-Platform Compatibility:** MySQL is compatible with multiple operating systems, including Windows, macOS, Linux, and various Unix variants. This flexibility allows developers to deploy MySQL on a wide range of platforms.
- **Scalability and Performance:** MySQL is designed to be highly scalable, capable of handling large volumes of data and high traffic loads. It offers features such as replication, sharding, and clustering to distribute data and workload across multiple servers for improved performance and reliability.
- **Data Security:** MySQL provides robust security features to protect data integrity and confidentiality. This includes user authentication, access control, encryption, and auditing capabilities to safeguard sensitive information from unauthorized access and attacks.
- **High Availability**: MySQL supports various high availability solutions, such as master-slave replication, multi-master replication, and clustering, to ensure continuous availability and fault tolerance. These features help minimize downtime and maintain data consistency in case of hardware failures or network issues.
- **Community and Support:** MySQL has a large and active community of developers, users, and contributors who provide support, share knowledge, and contribute to the improvement of the software. Additionally, commercial support and consulting services are available from MySQL vendors for organizations that require professional assistance.
- **Integration and Compatibility:** MySQL integrates seamlessly with popular web development technologies and frameworks, including PHP, Python, Ruby on Rails, Node.js, and Java. It also supports standard database interfaces such as ODBC and JDBC, enabling interoperability with a wide range of applications and tools.

Overall, MySQL is a powerful and versatile database management system that is well-suited for a variety of applications, from small-scale websites to large enterprise systems. Its combination of performance, scalability, reliability, and ease of use makes it a popular choice among developers and organizations worldwide.

5.3.2 How MySQL Works

MySQL works as a relational database management system (RDBMS) that stores and manages structured data organized into tables. Here's an overview of how MySQL works**:**

- Data Organization: MySQL organizes data into tables, where each table consists of rows and columns. Each column represents a specific attribute of the data, while each row contains a single record or tuple with values for those attributes.
- SQL Interface: MySQL uses SQL (Structured Query Language) as its interface for interacting with the database. SQL allows users to perform various operations such as querying data, inserting new records, updating existing records, and deleting records**.**
- **Client-Server Architecture:** MySQL follows a client-server architecture, where multiple clients can connect to a MySQL server to access and manipulate data. The MySQL server manages database operations and handles requests from clients, executing SQL queries and returning results as requested.
- **Connection Handling:** Clients connect to the MySQL server over a network using /IP or Unix sockets. Each client connection is authenticated using a username and password, and the server enforces access control based on user privileges and permissions.

- **Storage Engines:** MySQL supports multiple storage engines, each optimized for different types of workloads and data access patterns. The default storage engine for MySQL is InnoDB, which provides features such as ACID (Atomicity, Consistency, Isolation, Durability) transactions, foreign key constraints, and crash recovery. Overall, MySQL works by providing a reliable, efficient, and scalable platform for storing, managing, and accessing structured data, making it suitable for a wide range of applications and use cases.

5.3.3 MySQL Features

MySQL offers a rich set of features that make it a popular choice for building database-driven applications. Here are some key features of

MySQL:

- **Relational Database Management System (RDBMS)**: MySQL is a full- featured relational database management system that allows users to organize and manage structured data in tables with rows and columns. It supports SQL (Structured Query Language) for querying and manipulating data.
- **Cross-Platform Compatibility:** MySQL is compatible with various operating systems, including Windows, macOS, Linux, and Unix variants. This allows developers to deploy MySQL on a wide range of platforms and environments.
- **Scalability:** MySQL is designed to scale from small, single-server deployments to large, distributed systems handling millions of transactions per second. It supports features such as replication, sharding, and clustering to distribute data and workload across multiple servers for improved performance and scalability.
- **High Availability:** MySQL offers several features for ensuring high availability and fault tolerance, including master-slave replication, multi- master replication, and clustering. These features help minimize downtime and ensure continuous availability of data and services.
- **Security:** MySQL provides robust security features to protect data integrity and confidentiality. This includes user authentication, access control, encryption, and auditing capabilities to safeguard sensitive information from unauthorized access and attacks.

- **Performance Optimization:** MySQL includes various performance optimization features to improve query performance and throughput. This includes indexing, query caching, query optimization, and support for advanced storage engines such as InnoDB and MyISAM.
- **Data Types:** MySQL supports a wide range of data types, including numeric, string, date and time, spatial, and JSON data types. This allows developers to store and manipulate different types of data efficiently.
- **Stored Procedures and Functions:** MySQL supports stored procedures and user-defined functions, allowing developers to encapsulate SQL logic into reusable code blocks. This improves code modularity, maintainability, and performance.
- **Transaction Support:** MySQL provides support for transactions, which are sequences of SQL statements that are executed as a single unit of work. Transactions ensure that database changes are atomic, consistent, isolated, and durable (ACID properties), even in the event of system failures or interruptions.
- **Backup and Recovery:** MySQL includes features for backup and recovery, allowing users to create full or incremental backups of databases and restore them in the event of data loss or corruption.

Overall, MySQL offers a comprehensive set of features for building scalable, high-performance, and secure database-driven applications, making it a popular choice for developers and organizations worldwide.

5.4 Why use MySQL

There are several reasons why MySQL is a popular choice for building database-driven applications:

- **Performance:** MySQL is known for its high performance and efficiency, making it well- suited for applications that require fast and responsive data access. It supports various optimization techniques such as indexing, caching, and query optimization to improve query performance and throughput.
- **Scalability:** MySQL is highly scalable and can handle large volumes of data and high traffic loads. It supports features like replication, sharding, and clustering to

distribute data and workload across multiple servers, allowing applications to scale horizontally as demand grows.

- **Reliability**: MySQL is a stable and reliable database management system with a proven track record in production environments. It provides features for ensuring data integrity, transaction consistency, and fault tolerance, making it suitable for mission-critical applications.
- **Cross-Platform Compatibility**: MySQL is compatible with various operating systems and platforms, including Windows, macOS, Linux, and Unix variants. This allows developers to deploy MySQL on different environments without significant modifications to their applications.
- **Integration:** MySQL integrates seamlessly with popular programming languages, frameworks, and tools used in web development, such as PHP, Python, Ruby on Rails, Node.js, and Java. It also supports standard database interfaces such as ODBC and JDBC, enabling interoperability with a wide range of applications and tools.
- **Security**: MySQL provides robust security features to protect data integrity and confidentiality. This includes user authentication, access control, encryption, and auditing capabilities to safeguard sensitive information from unauthorized access and attacks.

Overall, MySQL offers a combination of performance, scalability, reliability, ease of use, and community support that makes it a preferred choice for building database-driven applications of all sizes and complexity levels

CHAPTER 6
SYSTEM ANALYSIS AND SYSTEM DESIGN

6.1 System Analysis and Design

System Analysis and Design (SAD) is the backbone of software development, guiding the creation of efficient and effective systems to address organizational needs. At its core, SAD is about understanding the current state of affairs within an organization, envisioning a better future, and architecting the pathway to get there.

During the analysis phase, SAD professionals delve deep into the existing systems and processes, meticulously identifying strengths, weaknesses, and opportunities for improvement. Through interviews, surveys, and observation, they gather requirements from stakeholders, ensuring that every voice is heard and every need is accounted for.

With requirements in hand, the design phase begins, where SAD experts transform abstract ideas into concrete plans. They define the system architecture, user interfaces, and data structures, ensuring that the design aligns closely with the gathered requirements and the overarching goals of the organization.

Implementation brings the design to life, as developers translate the blueprints into functional code. Rigorous testing follows, with each component scrutinized to ensure its reliability, security, and performance. User acceptance testing ensures that the system not only meets technical specifications but also satisfies the needs and expectations of its intended users.

Deployment marks the culmination of the SAD process, as the new system is introduced into the organization's ecosystem. With proper training and support, users adapt to the new system, embracing its capabilities and maximizing its potential to drive efficiency and innovation.

But the journey doesn't end there. Maintenance and support are ongoing responsibilities, as SAD professionals monitor the system, address issues, and incorporate updates to keep it running smoothly and meeting the evolving needs of the organization.

Throughout every phase of System Analysis and Design, collaboration, communication, and a relentless focus on the end-users are paramount. By aligning technology with business objectives and user needs, SAD professionals play a vital role in shaping the future success of organizations across industries.

6.2 Systems Development Life Cycle (SDLC)

A software life cycle model (also termed process model) is a pictorial and diagrammatic representation of the software life cycle. A life cycle model represents all the methods required to make a software product transit through its life cycle stages. It also captures the structure in which these methods are to be undertaken.

In other words, a life cycle model maps the various activities performed on a software product from its inception to retirement. Different life cycle models may plan the necessary development activities to phases in different ways. Thus, no element which life cycle model is followed, the essential activities are contained in all life cycle models though the action may be carried out in distinct orders in different life cycle models. During any life cycle stage, more than one activity may also be carried out.

6.2.1 Needs of SDLC

The development team must determine a suitable life cycle model for a particular plan and then observe to it.

Without using an exact life cycle model, the development of a software product would not be in a systematic and disciplined manner. When a team is developing a software product, there must be a clear understanding among team representative about when and what to do. Otherwise, it would point to chaos and project failure. This problem can be defined by using an example. Suppose a software development issue is divided into various parts and the parts are assigned to the team members. From then on, suppose the team representative is allowed the freedom to develop the roles assigned to them in whatever way they like. It is possible that one representative might start writing the code for his part, another might choose to prepare the test documents first, and some other engineer might begin with the design phase of the roles assigned to him. This would be one of the perfect methods for project failure.

A software life cycle model describes entry and exit criteria for each phase. A phase can begin only if its stage-entry criteria have been fulfilled. So without a software life cycle model, the entry and exit criteria for a stage cannot be recognized. Without software life cycle models, it becomes tough for software

project managers to monitor the progress of the project.

6.2.2 SDLC Cycle

SDLC Cycle represents the process of developing software. SDLC framework includes the following steps:

Figure 6.2.1: SDCL Cycle

- The stages of SDLC are as follows:

Stage1: Planning and requirement analysis

- Requirement Analysis is the most important and necessary stage in SDLC.
- The senior members of the team perform it with inputs from all the stakeholders and domain experts or SMEs in the industry.
- Planning for the quality assurance requirements and identifications of the risks associated with the projects is also done at this stage.

Stage2: Defining Requirements

- Once the requirement analysis is done, the next stage is to certainly represent and document the software requirements and get them accepted from the project stakeholders.
- This is accomplished through "SRS"- Software Requirement Specification document which contains all the product requirements to be constructed and developed during the project life cycle.

Stage3: Designing the Software

- The next phase is about to bring down all the knowledge of requirements, analysis, and design of the software project. This phase is the product of the last two, like inputs from the customer and requirement gathering.

Stage4: Developing the project

- In this phase of SDLC, the actual development begins, and the programming is built. The implementation of design begins concerning writing code.
- Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

Stage5: Testing

- After the code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage.
- During this stage, unit testing, integration testing, system testing, acceptance testing are done.

Stage6: Deployment

- Once the software is certified, and no bugs or errors are stated, then it is deployed.
- Then based on the assessment, the software may be released as it is or with suggested enhancement in the object segment. After the software is deployed, then its maintenance begins.

Stage7: Maintenance

- Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time. This procedure where the care is taken for the developed product is known as maintenance.

6.3 Spiral Model

The Spiral Model is a Software Development Life Cycle (SDLC) model that provides a systematic and iterative approach to software development. In its diagrammatic representation, looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development process.

The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks.

As the project manager dynamically determines the number of phases, the project manager has an important role in developing a product using the spiral model.

It is based on the idea of a spiral, with each iteration of the spiral representing a complete software development cycle, from analysis to design, implementation, testing, and maintenance.

The Spiral Model is a risk-driven model, meaning that the focus is on managing risk through multiple iterations of the software development process. **It consists of the following phases:**

- **Planning** – Understanding caregiver needs, clinician expectations, and defining system functionality.
- **Risk Analysis** – Identifying challenges like data security, user accessibility, and content authenticity.
- **Development & Testing –** Implementing features in cycles, testing for usability and security.
- **Evaluation & Refinement –** Gathering feedback, improving the system, and repeating the cycle.

c) Why Spiral Model for This Project?

- Allows for continuous improvement based on real-world usage.
- Helps manage risks early by addressing user needs iteratively.
- Supports scalability for adding new resources and features in future updates.

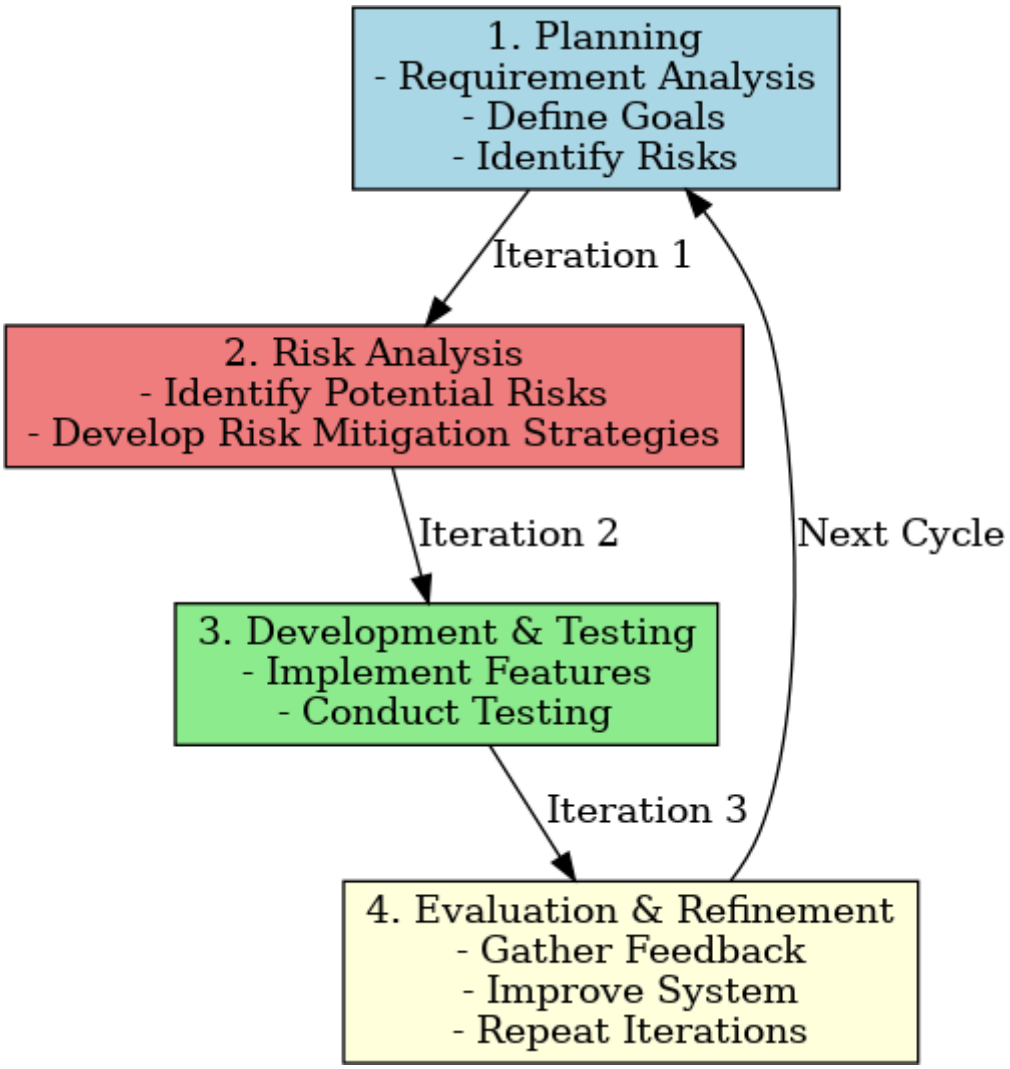


Figure 6.3.1: Spiral Model

6.4Class Diagram

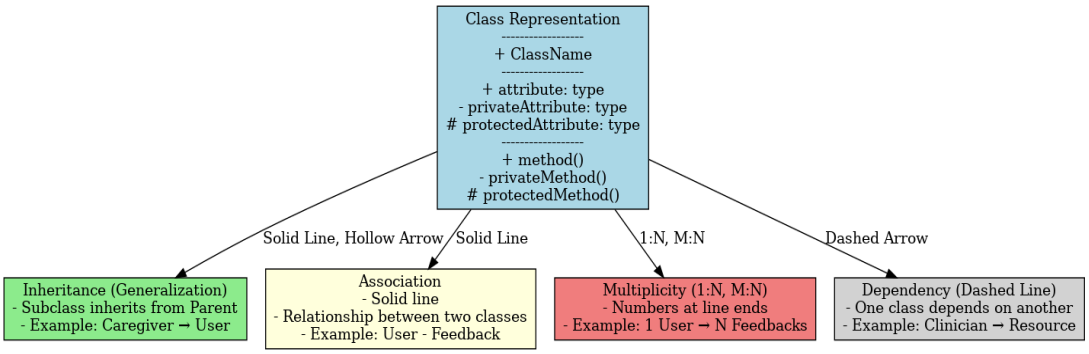
The class diagram depicts a static view of an application. It represents the types of objects residing in the system and the relationships between them. A class consists of its objects, and also it may inherit from other classes. A class diagram is used to visualize, describe, document various different aspects of the system, and also construct executable software code.

It shows the attributes, classes, functions, and relationships to give an overview of the software system. It constitutes class names, attributes, and functions in a separate compartment that helps in software development. Since it is a collection of classes, interfaces, associations, collaborations, and constraints, it is termed as a structural diagram.

- **Purpose of Class Diagrams**
  - The main purpose of class diagrams is to build a static view of an application. It is the only diagram that is widely used for construction, and it can be mapped with object-oriented languages. It is one of the most popular UML diagrams. Following are the purpose of class diagrams given below:
    - It analyses and designs a static view of an application.
    - It describes the major responsibilities of a system.
    - It is a base for component and deployment diagrams.
    - It incorporates forward and reverse engineering.
- **Benefits of Class Diagrams**
  - It can represent the object model for complex systems.
  - It reduces the maintenance time by providing an overview of how an application is structured before coding.
  - It provides a general schematic of an application for better understanding.

- It represents a detailed chart by highlighting the desired code, which is to be programmed.
- It is helpful for the stakeholders and the developers.

- **Symbols used in Class:**



CHAPTER: 7

DATA DICTIONARY

7.1 Introduction

Data Dictionary is an important part of a project or system which contains all definition of elements in the system. In Data Dictionary you will find a list of all elements composing the data flowing through a system.

- The major elements of a system:
  - Data Flow
  - Data Store
  - Processes
- The data dictionary stores all details and description of these elements.
- The data dictionary provides additional information about the system.
- The data dictionary contains the data about data e.g. Student is a data and this student belongs to this course is a description of STUDENT data which is stored in data dictionary.
- Why is Data Dictionary important?
  - To manage the details in large system.
  - To communicate a common meaning for all system elements.
  - To document the features of system.
  - To determine where system changes should made.
  - To locate errors and omissions in the system.

7.2 List of Tables:

- **Table 7.2.1: Admin_master**
  - **Description: This table stores data related to the admin module of the task management system.**

[Table 7.2.1: Admin_master]

Field Name	Data type (size)	Constraint	Description
admin_id	Integer	Primary key, Auto Increment	Unique identifier for the admin.
admin_name	Varchar(100)	Not Null	Full name of the admin.
admin_password	Varchar(32)	Not Null	Password for admin authentication.
admin_email	Varchar(100)	Not Null, Unique	Email address of the admin.

- **Table 7.2.2: User_ master**
  - **Description: This table stores data related to the user (employee) module of the task management system.**

[Table 7.2.2: User_ master]

Field Name	Data type (size)	Constraint	Description
user_id	Integer	Primary key, Auto Increment	Unique identifier of the user.
user_employee_id	Varchar(20)	Not Null, Unique	Employee ID of the user.
user_password	Varchar(32)	Not Null	Password of the user account.
user_fullname	Varchar(100)	Not Null	Full name of the user.
user_email	Varchar(100)	Not Null, Unique	Email address of the user.

- **Table 7.2.3: Task_ master**
  - **Description: This table stores information about tasks created in the task management system, including task details, priority, and status.**

[Table 7.2.3: Task _master]

Field Name	Data type (size)	Constraint	Description
task_id	Integer	Primary key, Auto Increment	Unique task identifier.
task_name	Varchar(100)	Not Null,unique	Name of the task.
task_description	text	Not Null	Description of the task.
task_status	Varchar(20)	Not Null	Task's current status.
task_priority	Varchar(20)	Not Null	Priority level of the task.

- **Table 7.2.4: Employee_Master**
  - **Description: This table stores information about tasks created in the task management system, including task details, priority, and status.**

[Table 7.2.4: Employee_Master]

Field Name	Data type (size)	Constraint	Description
emp_id	Integer	Primary key, Auto Increment	Unique employee identifier
emp_name	Varchar(100)	Not Null	Employee's full name.
emp_email	Varchar(100)	Not Null, Unique	Employee's email address.
emp_department	Varchar(50)	Not Null	Employee's department.
emp_designation	Varchar(50)	Not Null	Employee's job designation.

- **Table 7.2.5: Task_Assignment_Master**
  - **Description: This table tracks the assignment of tasks to employees, storing details like assigned employee, task ID, assigned date, and deadline.**

[Table 7.2.5: Task_Assignment_Master]

Field Name	Data type (size)	Constraint	Description
assignment_id	Integer	Primary key, Auto Increment	Unique task assignment identifier
emp_id	Integer	Foreign Key (Employee_Master)	Assigned employee ID.
task_id	Integer	Foreign Key (Task_Master)	Assigned task ID.
assigned_date	Date	Not Null	Task assignment date.
deadline	Date	Not Null	Task completion deadline

- **Table 7.2.6: Recent_Activity_Master**
  - **Description: This table logs user activities and actions performed in the task management system for tracking and monitoring purposes.**

[Table 7.2.6: Recent_Activity_Master]

Field Name	Data type (size)	Constraint	Description
activity_id	Integer	Primary key, Auto Increment	Unique activity identifier
user_id	Integer	Foreign Key (User_Master)	User performing the activity..
action	Varchar(255)	Not Null	Action performed by the user.
timestamp	timestamp	Default CURRENT_TIMESTAMP	Time of the activity.

- **Table 7.2.7: Project_Master**
  - **Description: This table stores project-related details, including project name, description, and timeline, for managing multiple tasks under a project.**

[Table 7.2.7: Project_Master]

Field Name	Data type (size)	Constraint	Description
project_id	Integer	Primary Key, Auto Increment	Unique project identifier.
project_name	Varchar(150)	Not Null, Unique	Name of the project.
project_description	Text	Not Null	Description of the project.
start_date	Date	Not Null	Project start date.
end_date	Date	Not Null	Project completion date.

CHAPTER:8

TESTING

8.1 Testing Plan

Great! Let's outline a detailed testing plan. Since you haven't specified the exact product or system you're testing, I'll provide a generic outline that you can adapt to your specific needs. Here's a structured testing plan:

- **Introduction:**
  - Provide an overview of the testing plan.
  - Include the objectives, scope, and key stakeholders.
- **Purpose:**
  - Define the purpose of testing.
  - Clarify what aspects of the product/system will be tested.
- **Scope:**
  - Describe the boundaries of the testing.
  - Identify the features, functionalities, or components included in the scope.
  - Mention any excluded items or functionalities.
- **Testing Goals:**
  - List the specific goals or objectives of the testing.
  - These goals should align with the overall project objectives and stakeholder requirements.
- **Testing Strategy:**
  - Outline the overall testing approach.
  - Specify the types of testing to be conducted (e.g., functional testing, usability testing, performance testing, security testing).
  - Describe any testing methodologies or frameworks to be followed (e.g., Agile, Waterfall, DevOps).
- **Test Deliverables:**
  - List the documents, reports, or artifacts that will be produced during testing.
  - Include items such as test plans, test cases, test scripts, and test reports.
- **Testing Environment:**
  - Specify the hardware, software, and network configurations required for testing.
  - Detail any test environment setup or configurations needed.

- **Test Cases:**
  - Develop detailed test cases for each identified test scenario.
  - Include preconditions, steps to execute, expected results, and post-conditions.
  - Organize test cases by priority and complexity.
- **Test Execution:**
  - Describe how test cases will be executed.
  - Assign responsibilities for test execution.
  - Define the criteria for determining when testing is complete.
- **Defect Management:**
  - Explain how defects/issues will be logged, tracked, and managed.
  - Define severity and priority levels for defects.
  - Outline the defect resolution process.
- **Risks and Mitigation:**
  - Identify potential risks associated with testing.
  - Propose mitigation strategies for each identified risk.
- **Schedule:**
  - Create a testing timeline or schedule.
  - Allocate time for test preparation, execution, and reporting.
  - Consider any dependencies or constraints that may impact the schedule.
- **Roles and Responsibilities:**
  - Define the roles and responsibilities of team members involved in testing.
  - Specify who is responsible for test planning, execution, documentation, and communication.
- **Communication Plan:**
  - Outline how communication will be managed throughout the testing process.
  - Identify key stakeholders and their communication preferences.
  - Specify the frequency and format of status updates and reports.
- **Review and Approval:**
  - Establish a process for reviewing and approving the testing plan.
  - Identify the stakeholders responsible for review and approval.
- **Documentation:**

  - Document any changes or updates to the testing plan.
  - Maintain a version history for reference.
- **Conclusion:**
  - Summarize the key points of the testing plan.
  - Confirm alignment with project goals and stakeholder expectations.
- **Appendices:**
  - Include any supplementary information or reference materials relevant to the testing plan.

Once you've drafted your testing plan, be sure to review it with your team and stakeholders to ensure clarity, alignment, and agreement before proceeding with testing activities.

8.2 Testing Strategies

To ensure the reliability and functionality of the Task Management System, the following testing strategies will be applied:

- **Unit Testing :**
  - Testing individual modules or components (like login, task creation) in isolation.
- **Integration Testing :**
  - Testing how different modules (e.g., task assignment and user details) interact with each other.
- **System Testing :**
  - Testing the entire application as a whole to ensure it meets the specified requirements.
- **User Acceptance Testing (UAT) :**
  - Testing from the end user's point of view to confirm that the system fulfills its intended use.
- **Regression Testing :**
  - Re-testing existing features after changes or bug fixes to ensure new code hasn't affected the existing system.

8.3 Testing Methods

- **Manual Testing**

- Testers manually check all functionalities such as login, task creation, project management, etc., to ensure expected behavior.

- **Black Box Testing**
  - Focuses on input and output of software applications. Internal code structure is not considered, useful for functional testing.

- **White Box Testing**
  - Focuses on internal logic and structure of the code. Useful for checking logical errors, loops, conditions, etc.

- **Automation Testing *(Optional/Advanced)***
  - Tools like Selenium can be used to automate repetitive tasks like login testing, form submissions, etc., for faster results

8.4 Test Case

A test case is a detailed set of conditions or steps that are designed to verify the functionality, behavior, or performance of a software application or system. Test cases are used to validate whether the software meets specific requirements and functions correctly under various scenarios. Here's a breakdown of what a test case typically includes:

- **Test Case ID:** A unique identifier for the test case, usually in the form of a combination of letters and numbers.
- **Test Case Title/Name:** A descriptive title or name that summarizes the purpose or objective of the test case.
- **Description:** A brief description or summary of what the test case is intended to achieve.
- **Preconditions:** User must be registered in the system with valid credentials. The login page must be accessible.

- **Test Steps:**
  - Navigate to the login page
  - Enter valid email

  - Enter valid Role
  - Click on “Subscribe” button

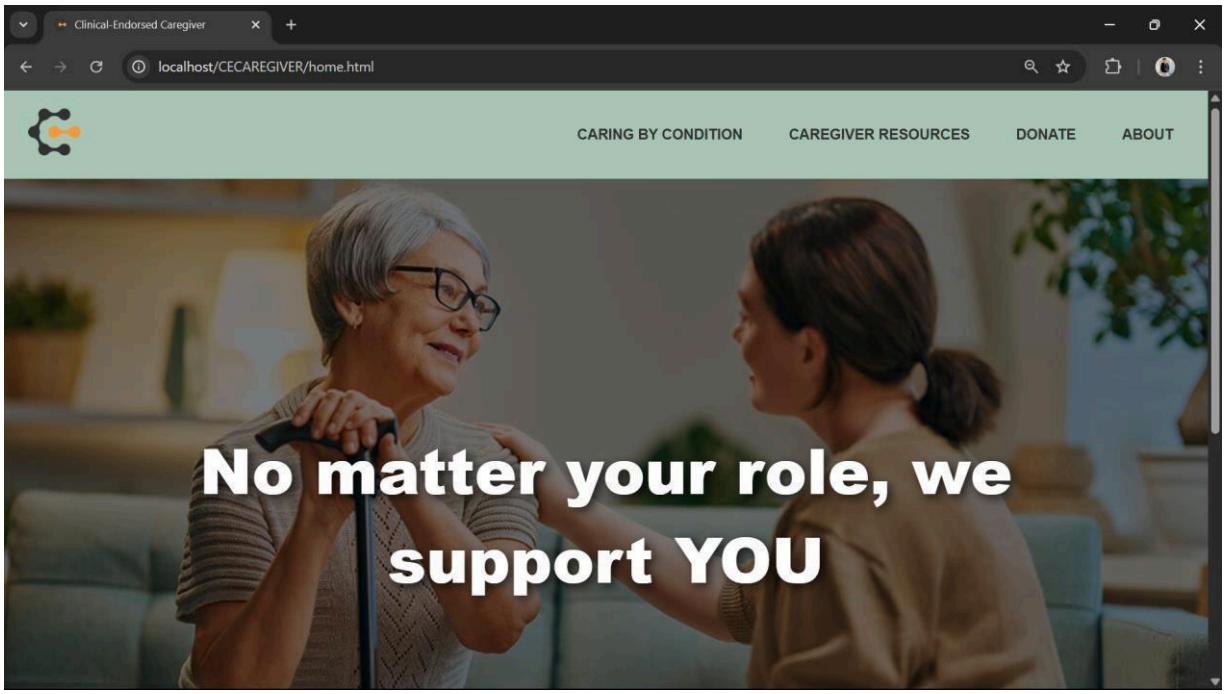
- **Input Data:** The specific data values, parameters, or inputs that will be used during the test execution.
- **Expected Results:** The expected outcomes or behaviors that the system should exhibit when the test case is executed successfully. This includes both observable results and system responses.
- **Actual Results:** The actual outcomes observed during test execution. Testers record the results obtained after executing the test steps.
- **Pass/Fail Criteria**: Criteria used to determine whether the test case passes or fails based on a comparison between the actual results and the expected results.
- **Priority:** The priority level assigned to the test case, indicating its relative importance or urgency in the testing process.
- **Severity:** The severity level assigned to the test case, indicating the impact of a failure on the system or its users. Common severity levels include critical, major, minor, and trivial.
- **Test Environment:** Information about the test environment, including hardware, software, configurations, and dependencies required for test execution.
- **Test Data:** Any additional test data or conditions relevant to the test case, such as boundary values, edge cases, or negative scenarios.
- **Attachments/References:** Any supporting documents, screenshots, logs, or references that provide additional context or details for the test case.
- **Notes/Comments**: Any additional notes, comments, or observations relevant to the test case, such as insights, assumptions, or recommendations.

By documenting test cases in a structured and standardized format, teams can ensure consistent testing practices, facilitate test execution, and maintain comprehensive test coverage throughout the software development lifecycle. Additionally, well-written test cases serve as valuable documentation for future reference, regression testing, and knowledge transfer within the team.

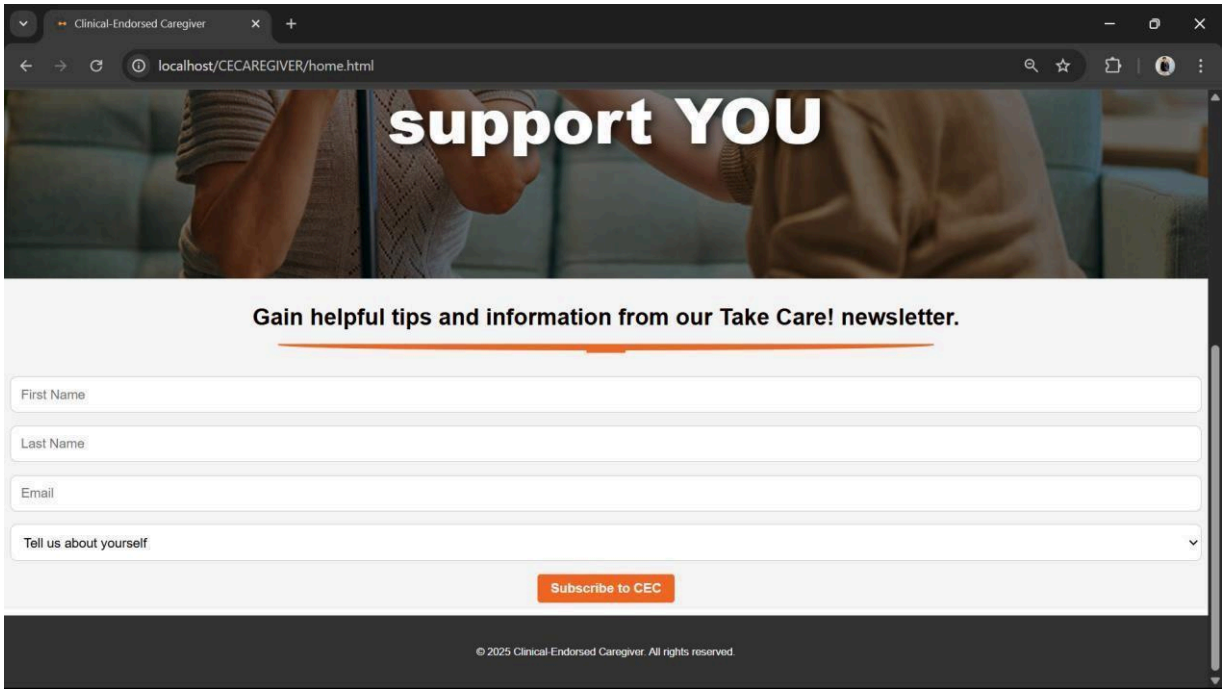
CHAPTER: 9

SNAPSHOT OF WEBSITE

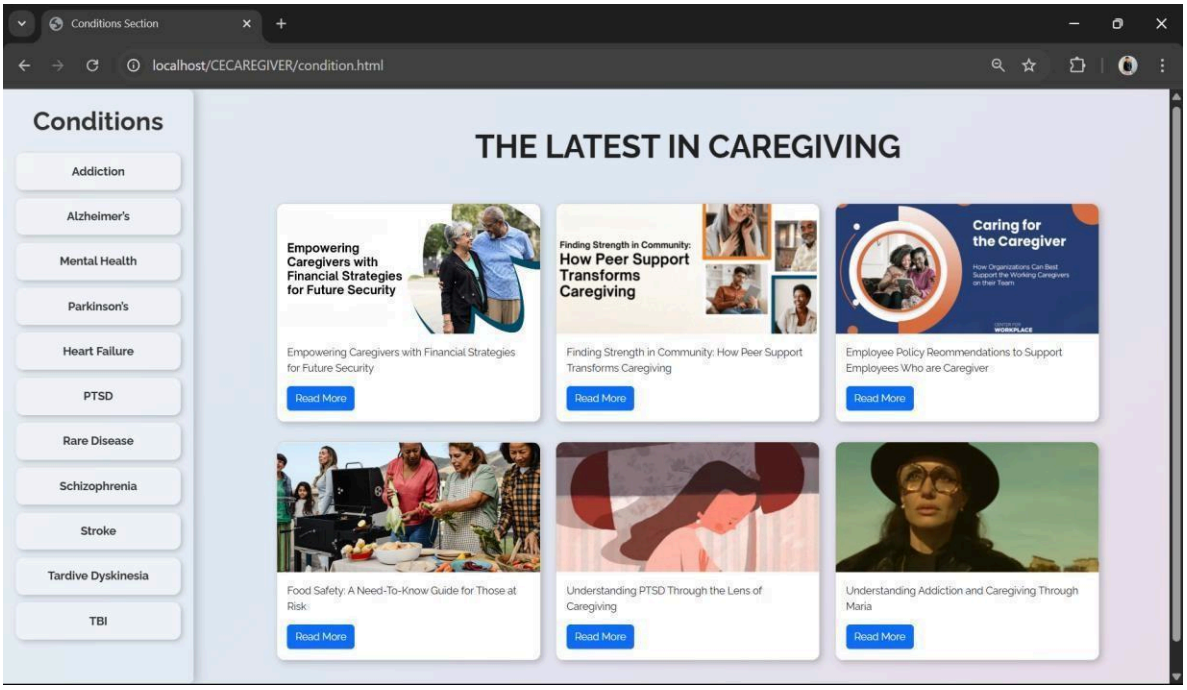
- **Home Page :**



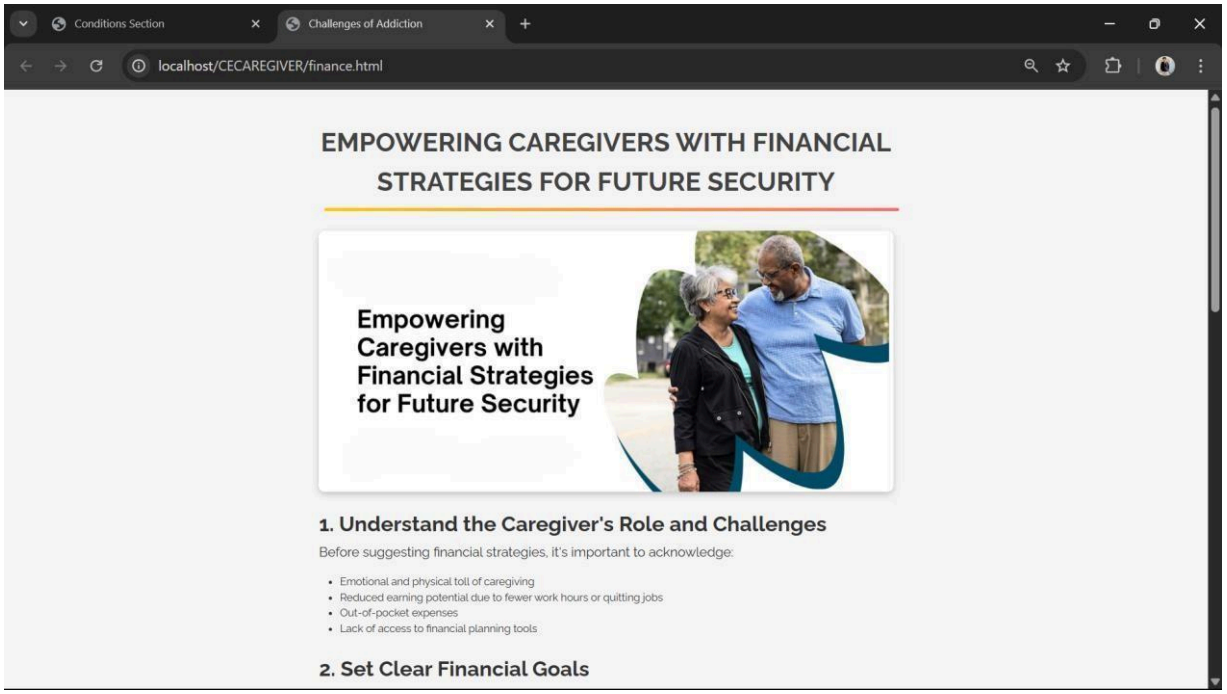
- **Login Page :**

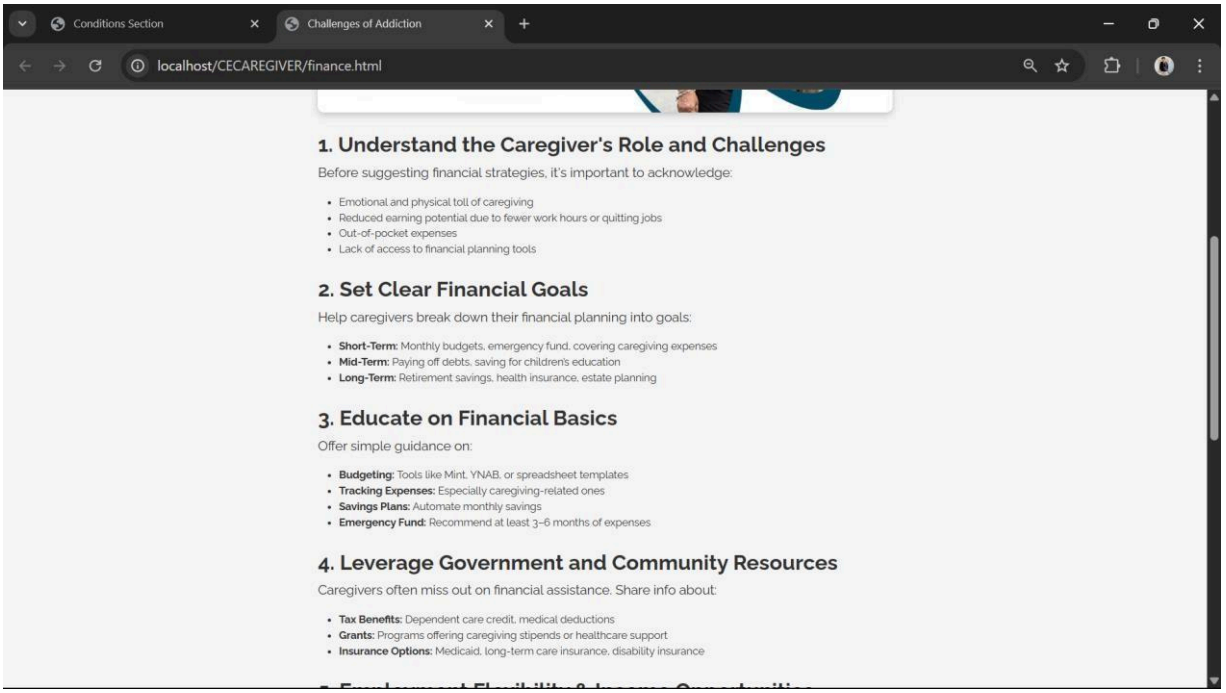


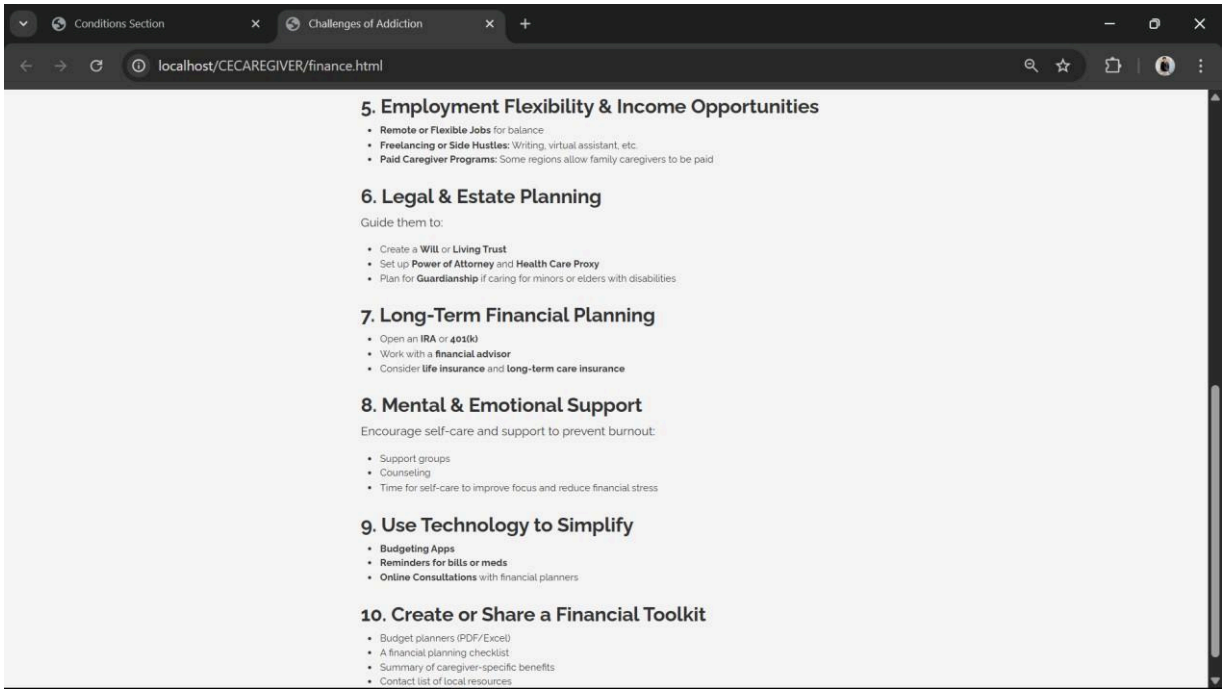
- **The Latest in Caregiving :**



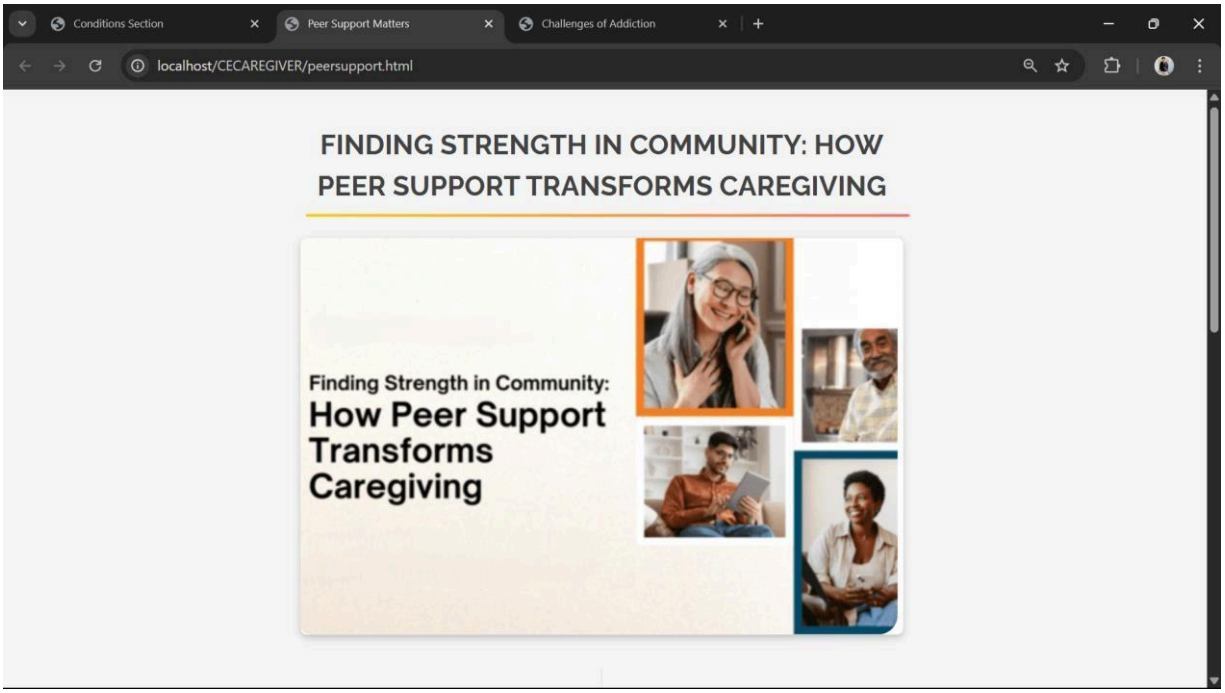
- **Empowering Caregivers with Financial Strategies for Future Security**

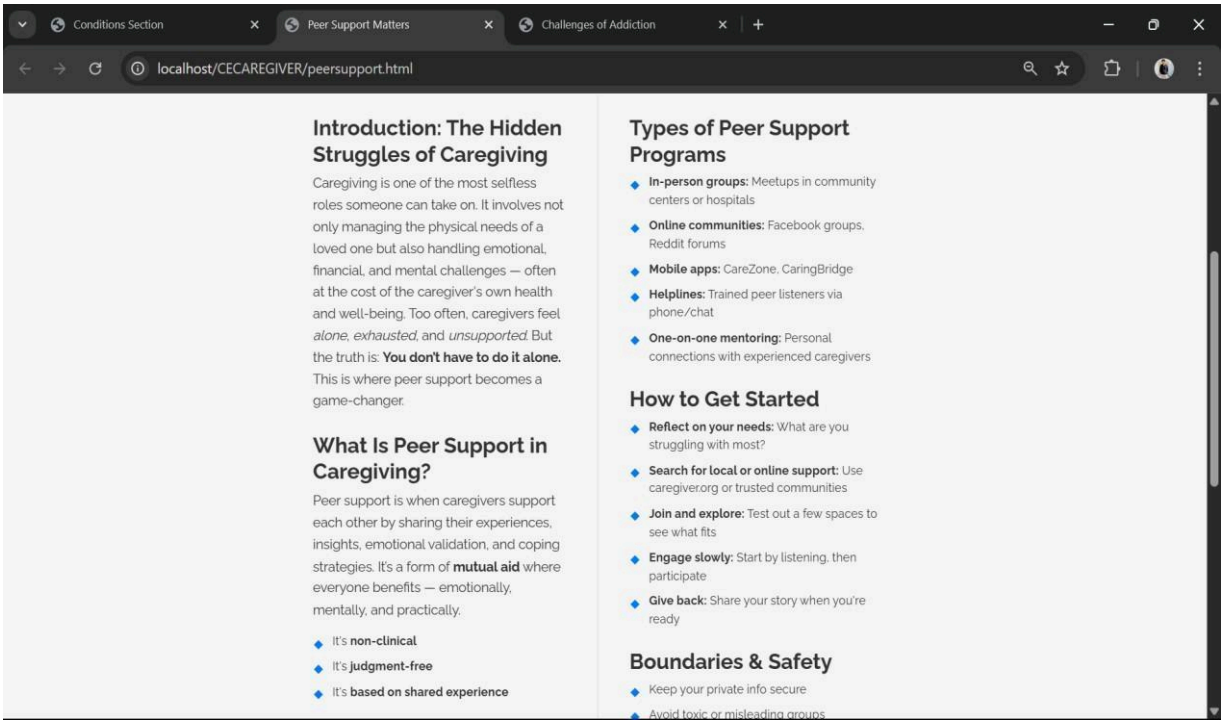


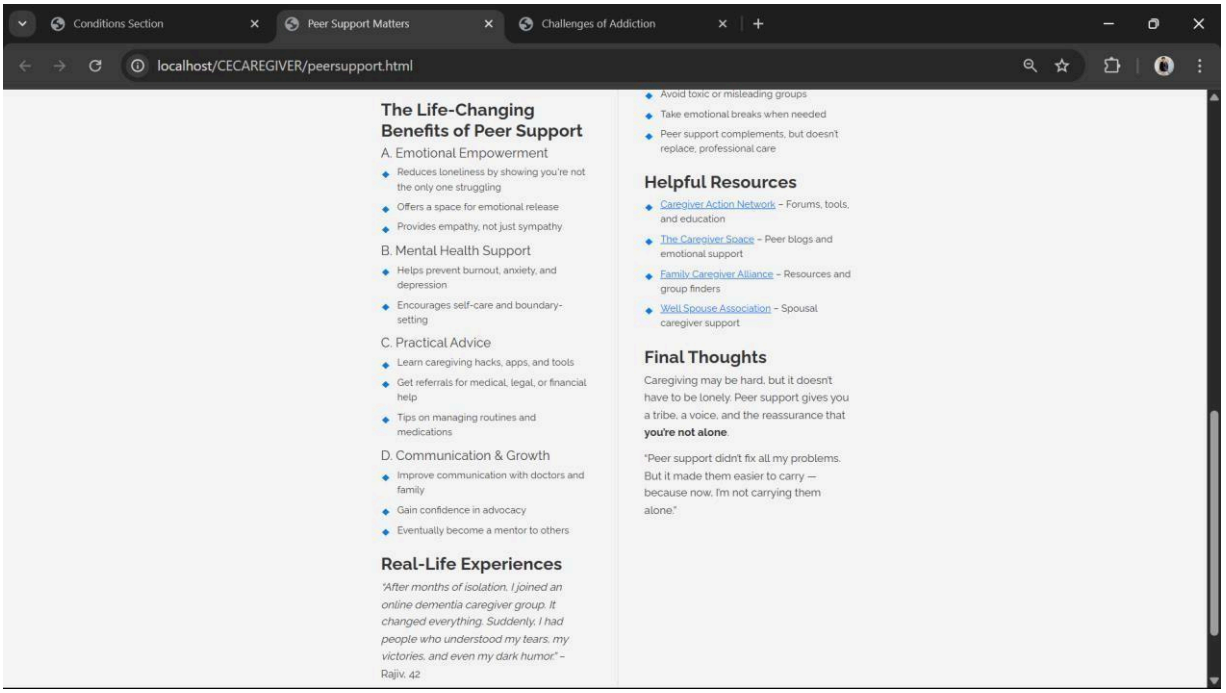

Conditions Section
Challenges of Addiction
localhost/CECAREGIVER/finance.html
1. Understand the Caregiver's Role and Challenges
Before suggesting financial strategies, it's important to acknowledge:
Emotional and physical toll of caregiving
Reduced earning potential due to fewer work hours or quitting jobs
Out-of-pocket expenses
Lack of access to financial planning tools
2. Set Clear Financial Goals
Help caregivers break down their financial planning into goals:
Short-Term: Monthly budgets, emergency fund, covering caregiving expenses
Mid-Term: Paying off debts, saving for children's education
Long-Term: Retirement savings, health insurance, estate planning
3. Educate on Financial Basics
Offer simple guidance on:
Budgeting: Tools like Mint, YNAB, or spreadsheet templates
Tracking Expenses: Especially caregiving-related ones
Savings Plans: Automate monthly savings
Emergency Fund: Recommend at least 3–6 months of expenses
4. Leverage Government and Community Resources
Caregivers often miss out on financial assistance. Share info about:
Tax Benefits: Dependent care credit, medical deductions
Grants: Programs offering caregiving stipends or healthcare support
Insurance Options: Medicaid, long-term care insurance, disability insurance


Conditions Section
Challenges of Addiction
localhost/CECAREGIVER/finance.html
5. Employment Flexibility & Income Opportunities
Remote or Flexible Jobs for balance
Freelancing or Side Hustles: Writing, virtual assistant, etc.
Paid Caregiver Programs: Some regions allow family caregivers to be paid
6. Legal & Estate Planning
Guide them to:
Create a Will or Living Trust
Set up Power of Attorney and Health Care Proxy
Plan for Guardianship if caring for minors or elders with disabilities
7. Long-Term Financial Planning
Open an IRA or 401(k)
Work with a financial advisor
Consider life insurance and long-term care insurance
8. Mental & Emotional Support
Encourage self-care and support to prevent burnout:
Support groups
Counseling
Time for self-care to improve focus and reduce financial stress
9. Use Technology to Simplify
Budgeting Apps
Reminders for bills or meds
Online Consultations with financial planners
10. Create or Share a Financial Toolkit
Budget planners (PDF/Excel)
A financial planning checklist
Summary of caregiver-specific benefits
Contact list of local resources

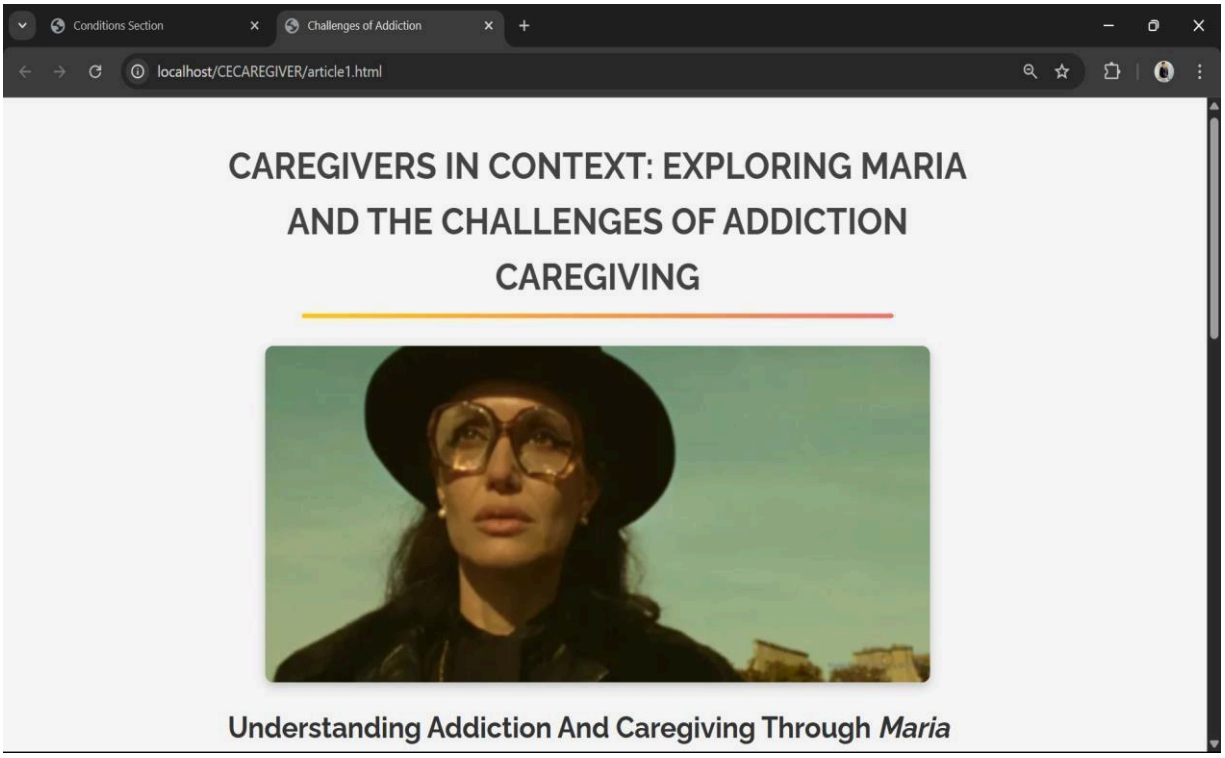
- **Finding Strength in Community: How Peer Support Transforms Caregiving :**

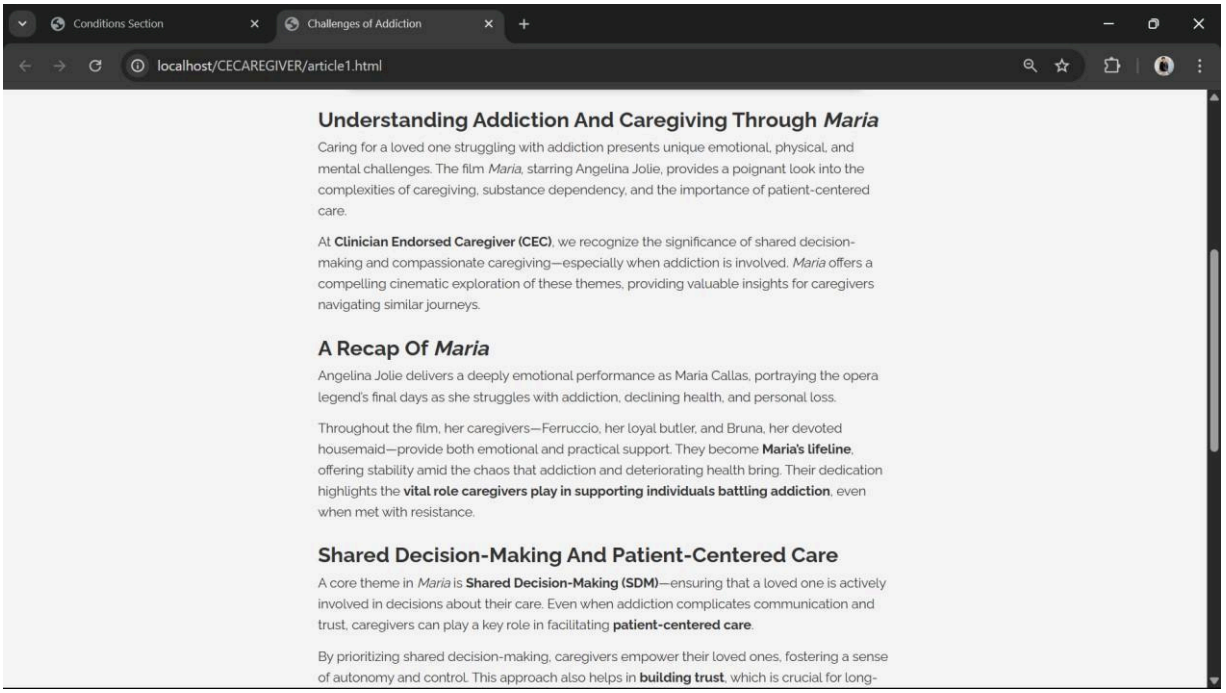


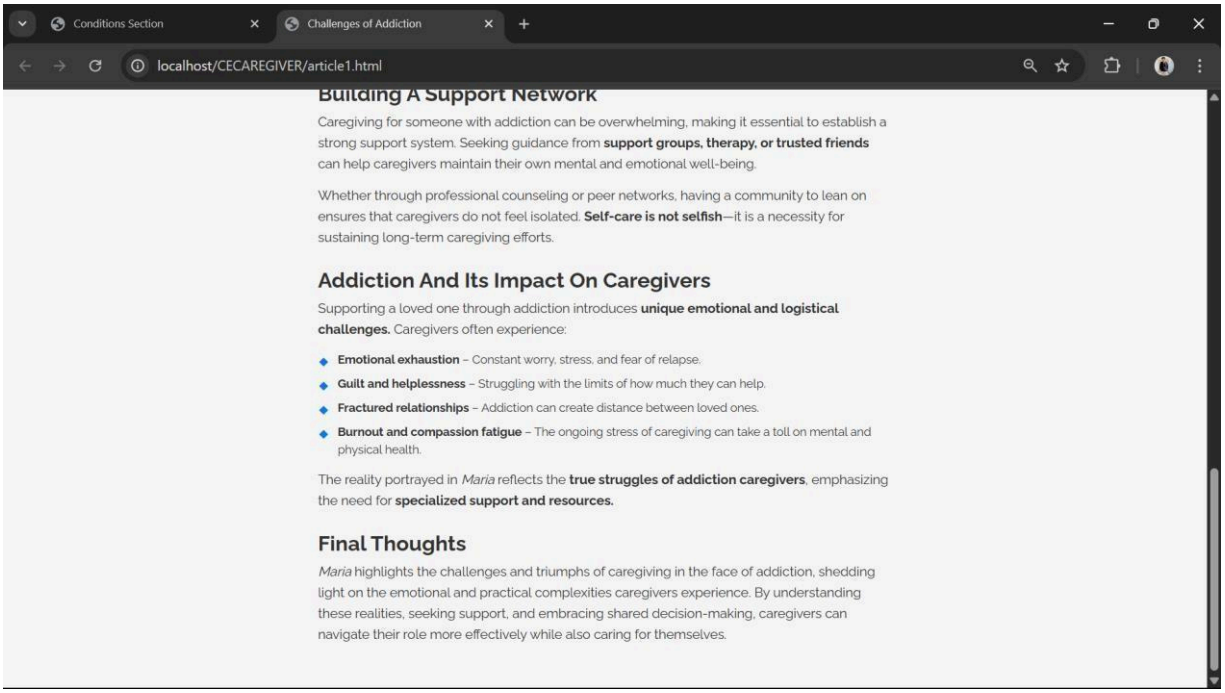




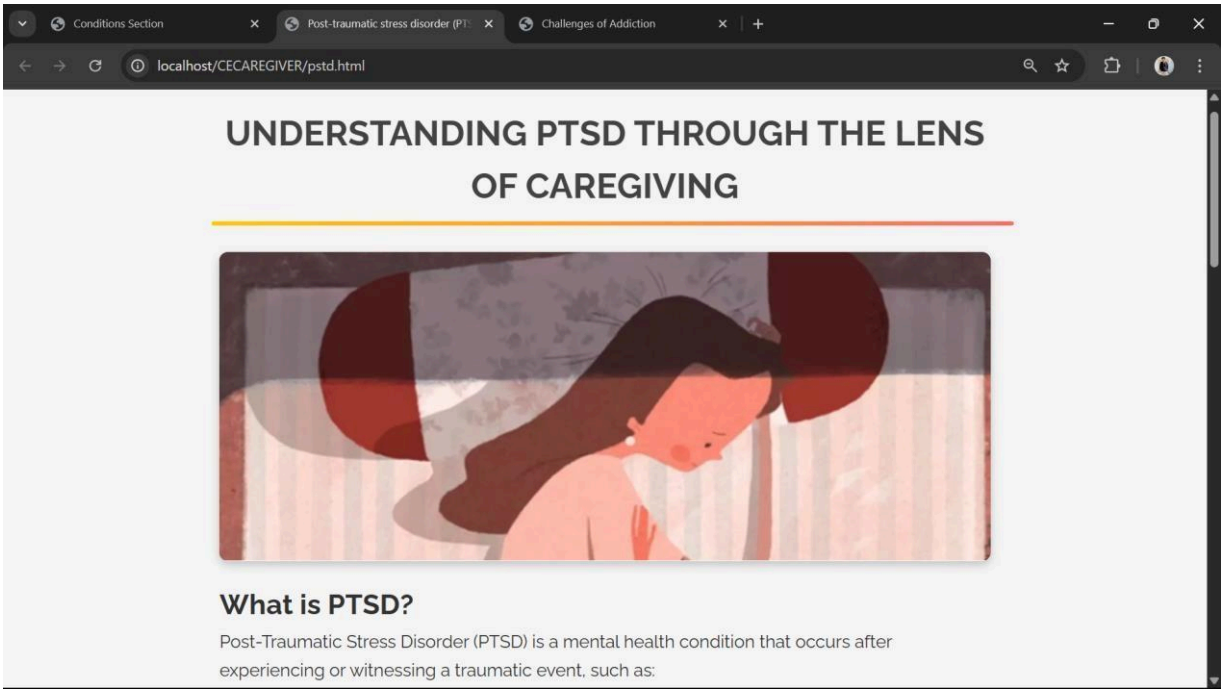
- **Caregivers In Context: Exploring Maria And The Challenges Of Addiction Caregiving :**



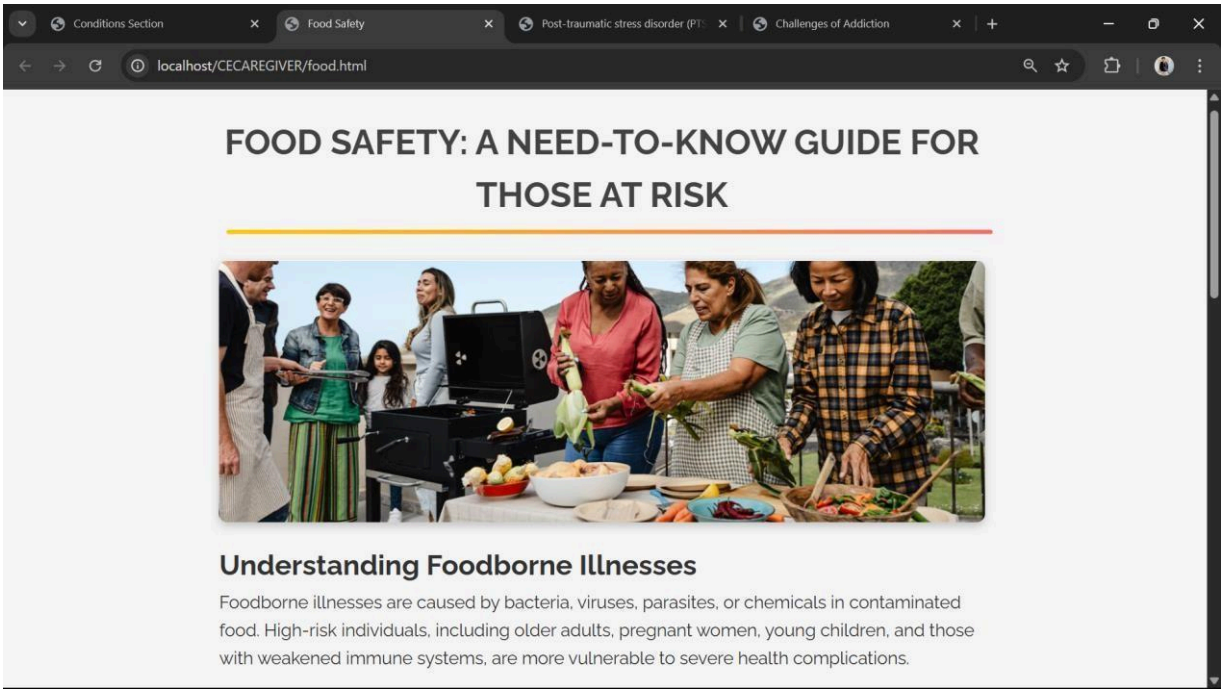

Conditions Section
Challenges of Addiction
localhost/CECAREGIVER/article1.html
Understanding Addiction And Caregiving Through Maria
Caring for a loved one struggling with addiction presents unique emotional, physical, and mental challenges. The film Maria, starring Angelina Jolie, provides a poignant look into the complexities of caregiving, substance dependency, and the importance of patient-centered care.
At Clinician Endorsed Caregiver (CEC), we recognize the significance of shared decision-making and compassionate caregiving—especially when addiction is involved. Maria offers a compelling cinematic exploration of these themes, providing valuable insights for caregivers navigating similar journeys.
A Recap Of Maria
Angelina Jolie delivers a deeply emotional performance as Maria Callas, portraying the opera legend's final days as she struggles with addiction, declining health, and personal loss.
Throughout the film, her caregivers—Ferruccio, her loyal butler, and Bruna, her devoted housemaid—provide both emotional and practical support. They become Maria's lifeline, offering stability amid the chaos that addiction and deteriorating health bring. Their dedication highlights the vital role caregivers play in supporting individuals battling addiction, even when met with resistance.
Shared Decision-Making And Patient-Centered Care
A core theme in Maria is Shared Decision-Making (SDM)—ensuring that a loved one is actively involved in decisions about their care. Even when addiction complicates communication and trust, caregivers can play a key role in facilitating patient-centered care.
By prioritizing shared decision-making, caregivers empower their loved ones, fostering a sense of autonomy and control. This approach also helps in building trust, which is crucial for long-


Conditions Section
Challenges of Addiction
localhost/CECAREGIVER/article1.html
Building A Support Network
Caregiving for someone with addiction can be overwhelming, making it essential to establish a strong support system. Seeking guidance from support groups, therapy, or trusted friends can help caregivers maintain their own mental and emotional well-being.
Whether through professional counseling or peer networks, having a community to lean on ensures that caregivers do not feel isolated. Self-care is not selfish—it is a necessity for sustaining long-term caregiving efforts.
Addiction And Its Impact On Caregivers
Supporting a loved one through addiction introduces unique emotional and logistical challenges. Caregivers often experience:
Emotional exhaustion – Constant worry, stress, and fear of relapse.
Guilt and helplessness – Struggling with the limits of how much they can help.
Fractured relationships – Addiction can create distance between loved ones.
Burnout and compassion fatigue – The ongoing stress of caregiving can take a toll on mental and physical health.
The reality portrayed in Maria reflects the true struggles of addiction caregivers, emphasizing the need for specialized support and resources.
Final Thoughts
Maria highlights the challenges and triumphs of caregiving in the face of addiction, shedding light on the emotional and practical complexities caregivers experience. By understanding these realities, seeking support, and embracing shared decision-making, caregivers can navigate their role more effectively while also caring for themselves.

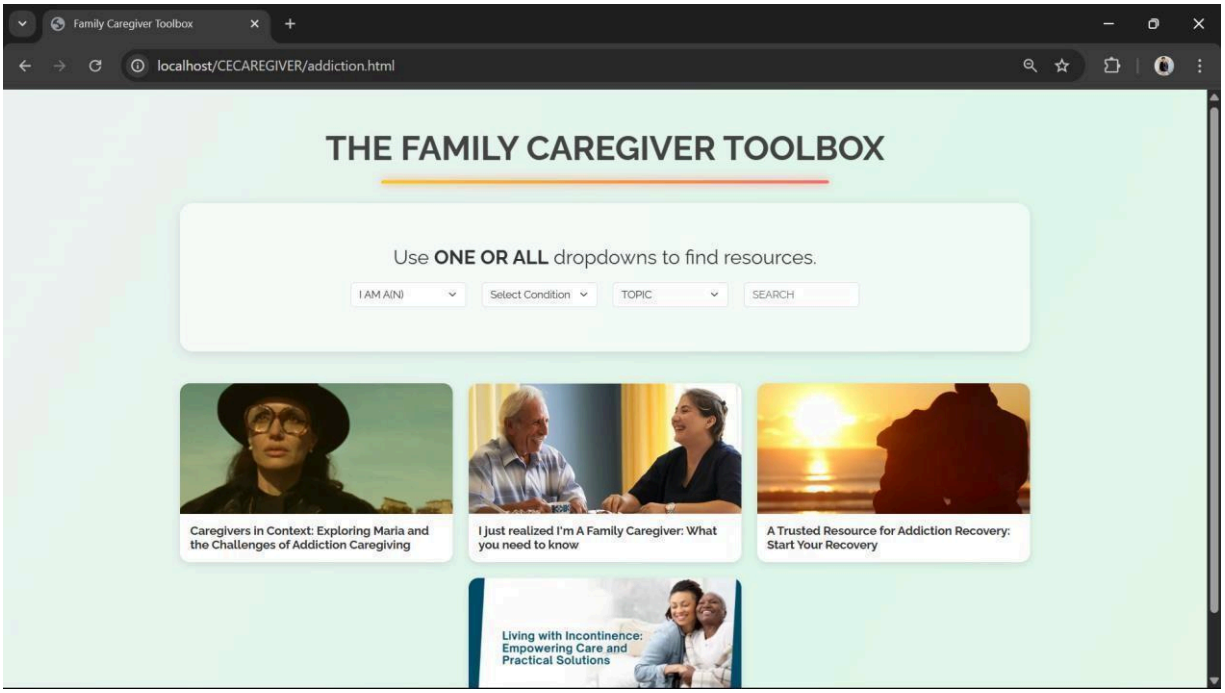
- **Understanding PTSD Through the Lens of Caregiving :**



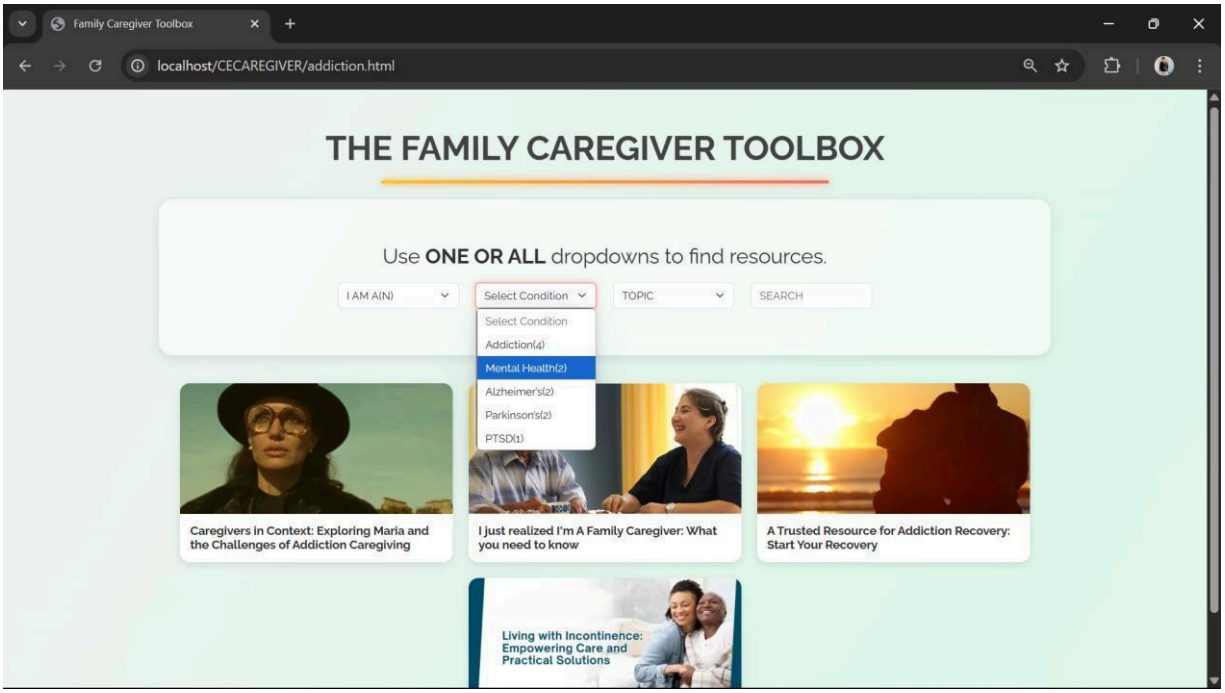
- **Food Safety: A Need-To-Know Guide for Those at Risk :**



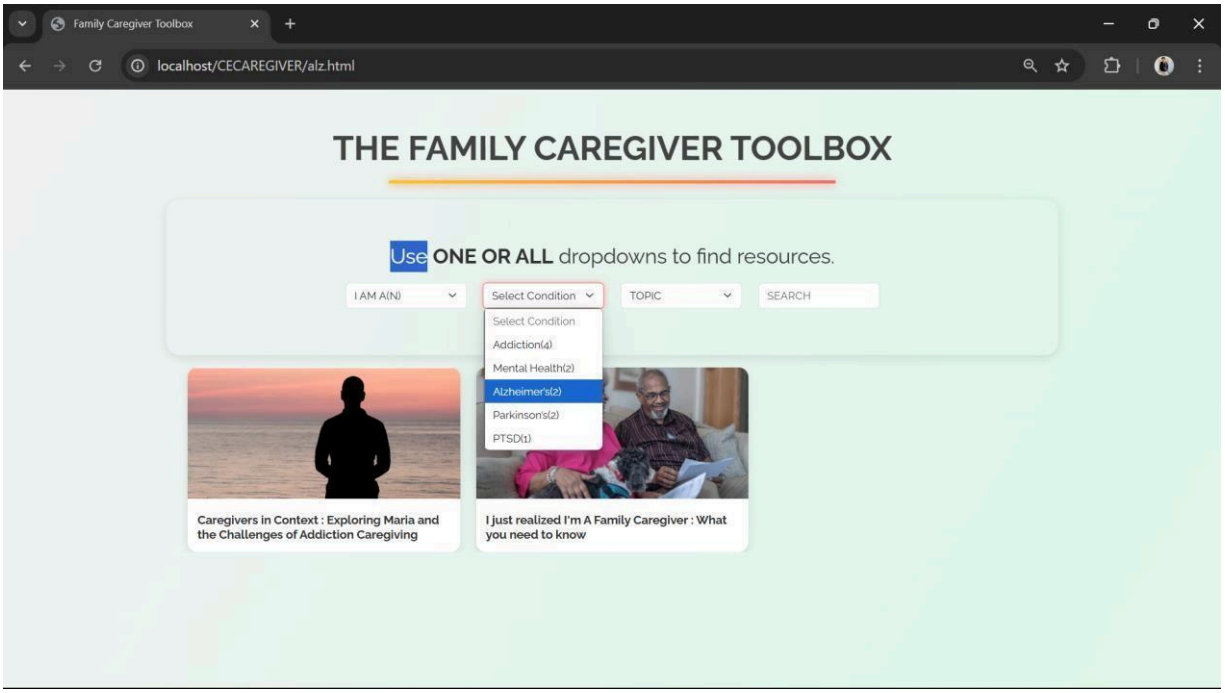
- **Addiction :**

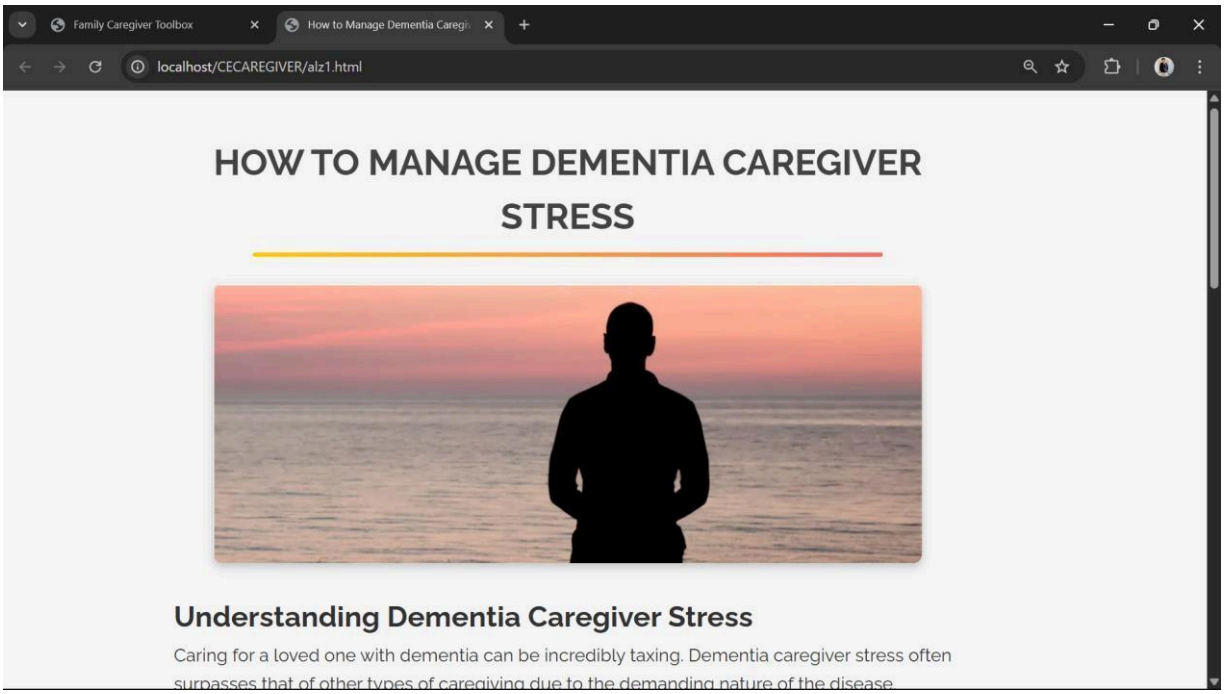


- **Select Conditions :**

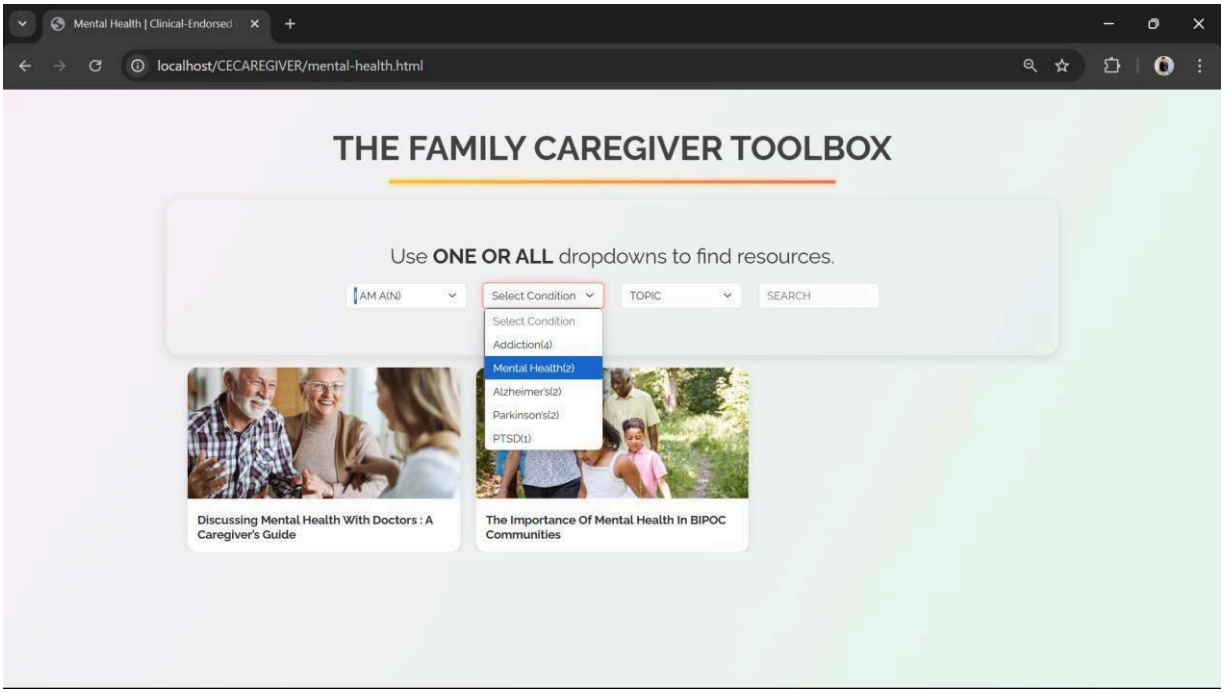


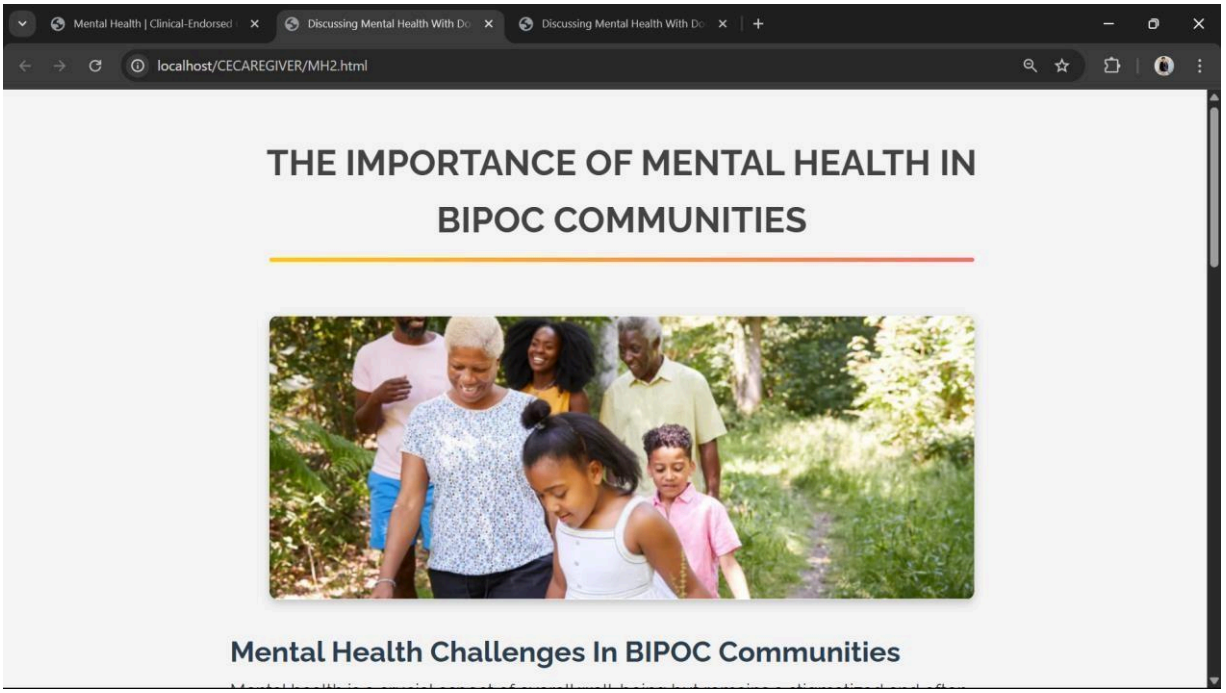
- **Alzheimer's :**


Family Caregiver Toolbox
localhost/CECAREGIVER/alz.html
THE FAMILY CAREGIVER TOOLBOX
Use ONE OR ALL dropdowns to find resources.
I AM A(N)
Select Condition
TOPIC
SEARCH
Select Condition
Addiction(4)
Mental Health(2)
Alzheimer's(2)
Parkinson's(2)
PTSD(1)
Caregivers in Context : Exploring Maria and the Challenges of Addiction Caregiving
I just realized I'm A Family Caregiver : What you need to know

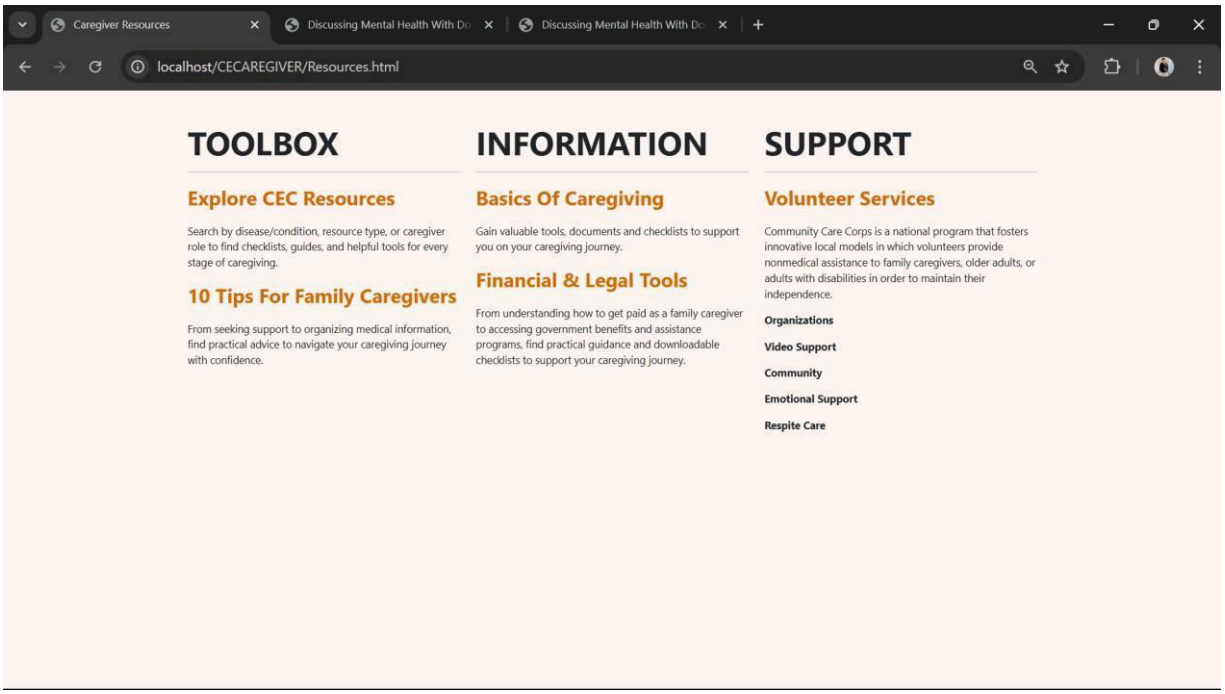

Family Caregiver Toolbox
How to Manage Dementia Caregi
localhost/CECAREGIVER/alz1.html
HOW TO MANAGE DEMENTIA CAREGIVER STRESS
Understanding Dementia Caregiver Stress
Caring for a loved one with dementia can be incredibly taxing. Dementia caregiver stress often surpasses that of other types of caregiving due to the demanding nature of the disease.

- **Mental Health :**

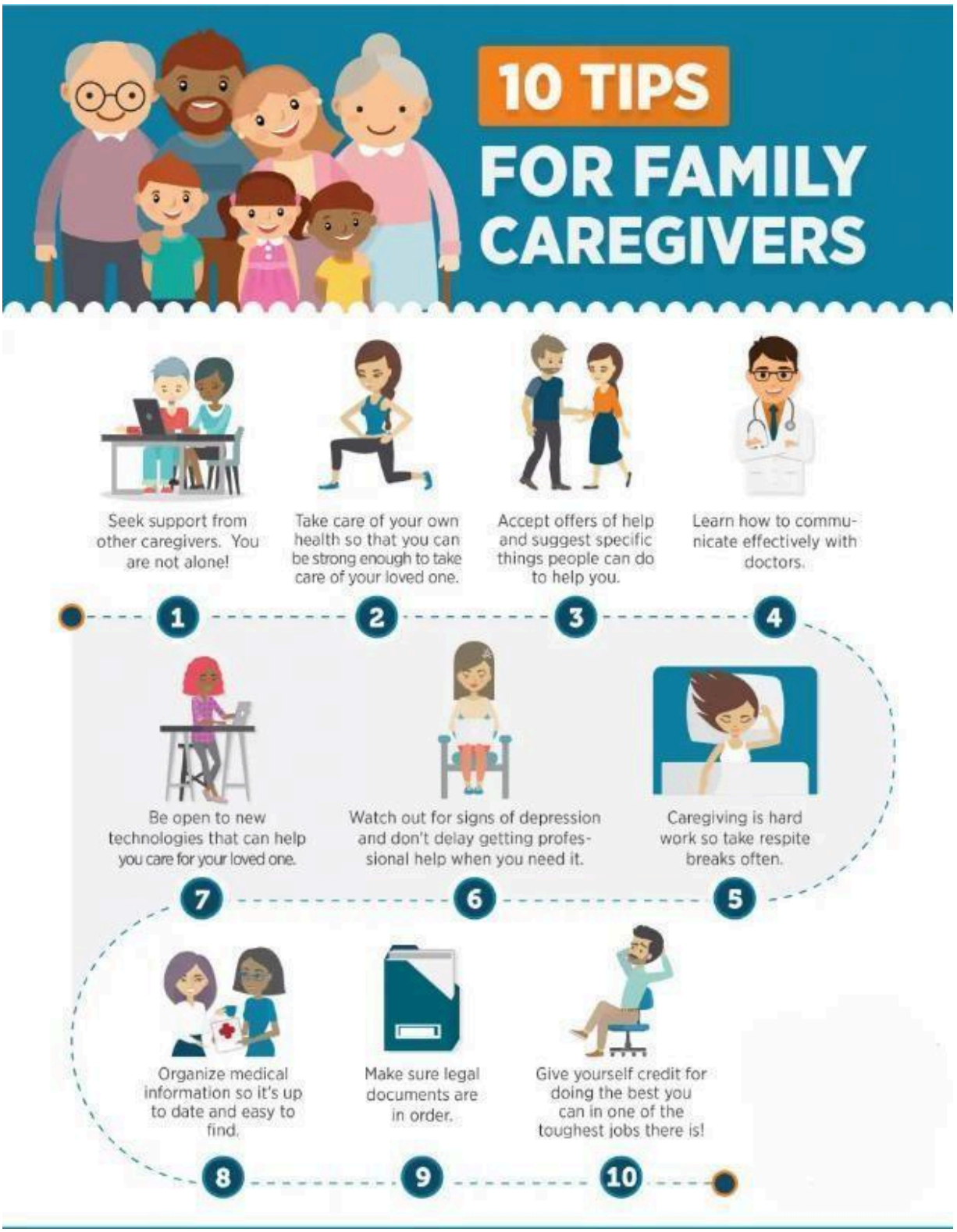




- **Caregiver Resources :**



- **Tips for Family Caregivers: Balancing Caregiving and Self-Care :**



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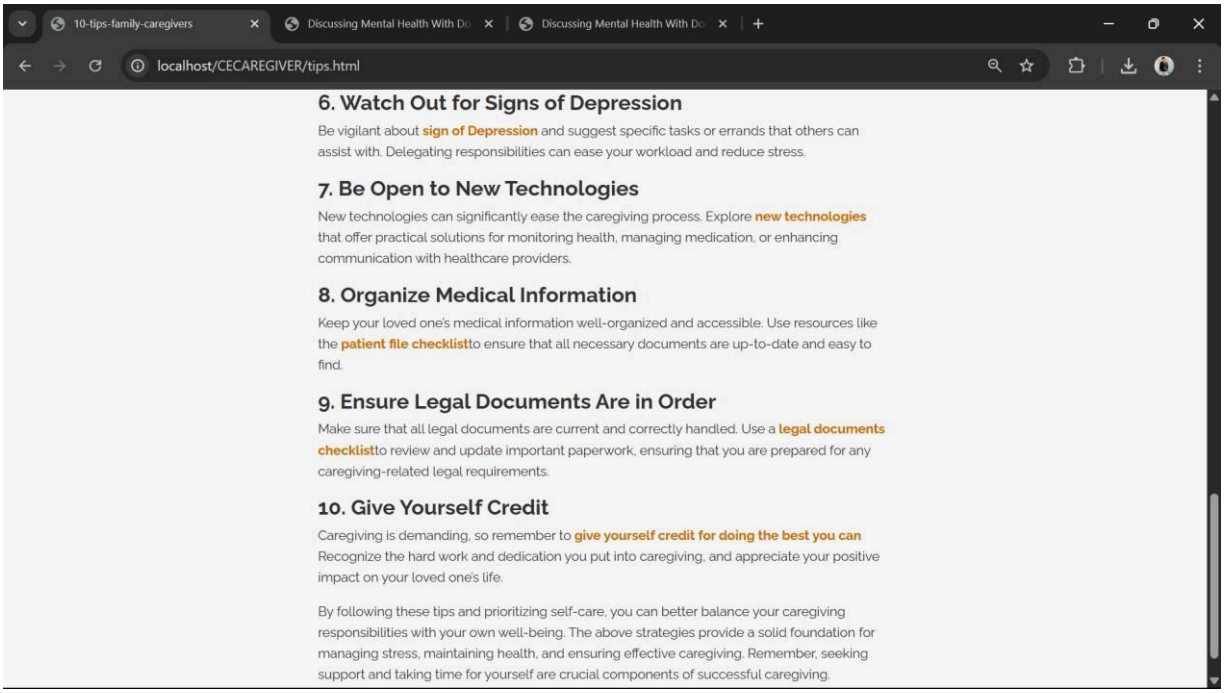
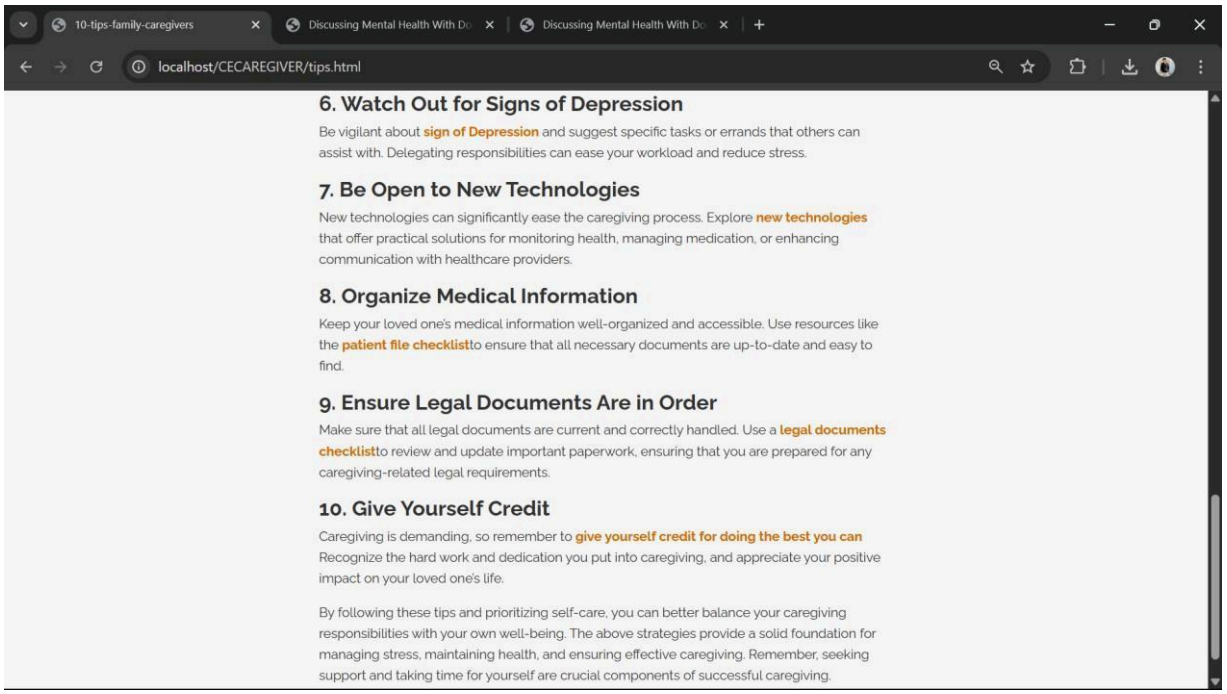

10-tips-family-caregivers
Discussing Mental Health With Do
Discussing Mental Health With Do
localhost/CECAREGIVER/tips.html
6. Watch Out for Signs of Depression
Be vigilant about sign of Depression and suggest specific tasks or errands that others can assist with. Delegating responsibilities can ease your workload and reduce stress.
7. Be Open to New Technologies
New technologies can significantly ease the caregiving process. Explore new technologies that offer practical solutions for monitoring health, managing medication, or enhancing communication with healthcare providers.
8. Organize Medical Information
Keep your loved one's medical information well-organized and accessible. Use resources like the patient file checklistto ensure that all necessary documents are up-to-date and easy to find.
9. Ensure Legal Documents Are in Order
Make sure that all legal documents are current and correctly handled. Use a legal documents checklistto review and update important paperwork, ensuring that you are prepared for any caregiving-related legal requirements.
10. Give Yourself Credit
Caregiving is demanding, so remember to give yourself credit for doing the best you can Recognize the hard work and dedication you put into caregiving, and appreciate your positive impact on your loved one's life.
By following these tips and prioritizing self-care, you can better balance your caregiving responsibilities with your own well-being. The above strategies provide a solid foundation for managing stress, maintaining health, and ensuring effective caregiving. Remember, seeking support and taking time for yourself are crucial components of successful caregiving.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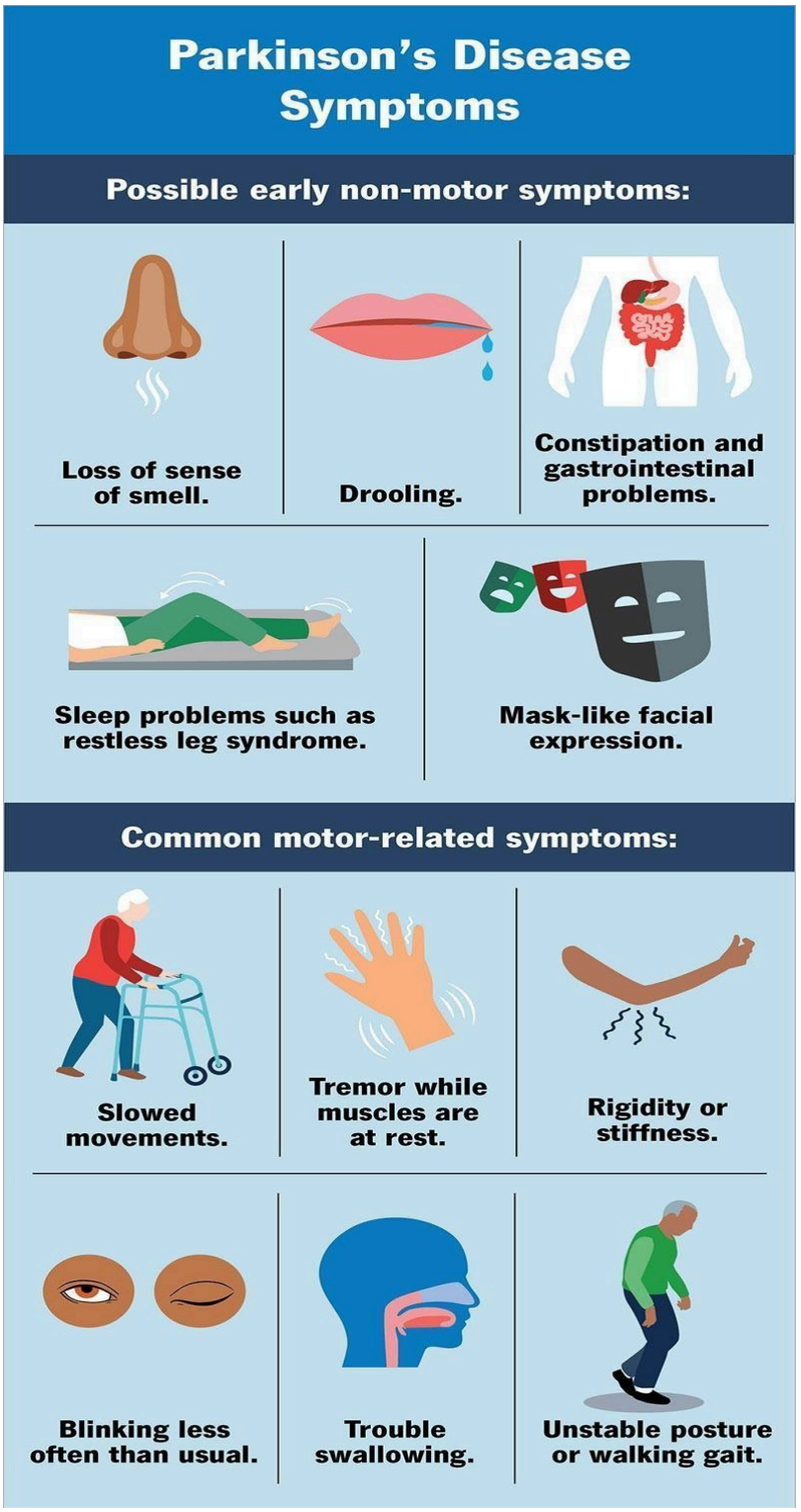
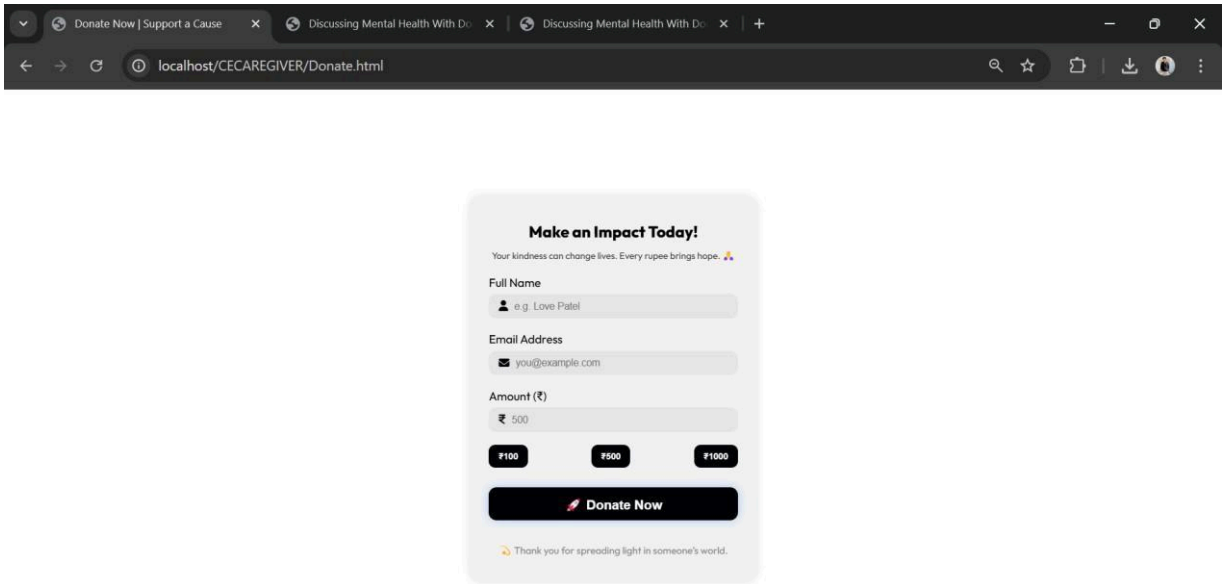
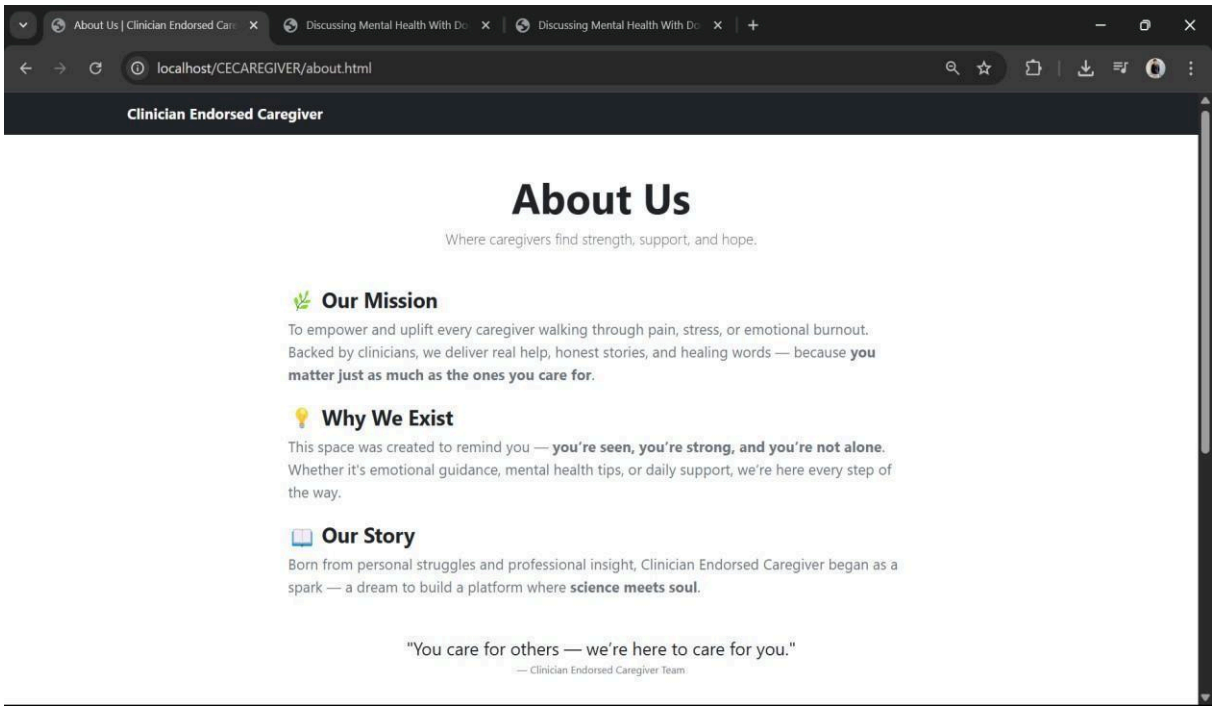
- **parkinsons-disease :**

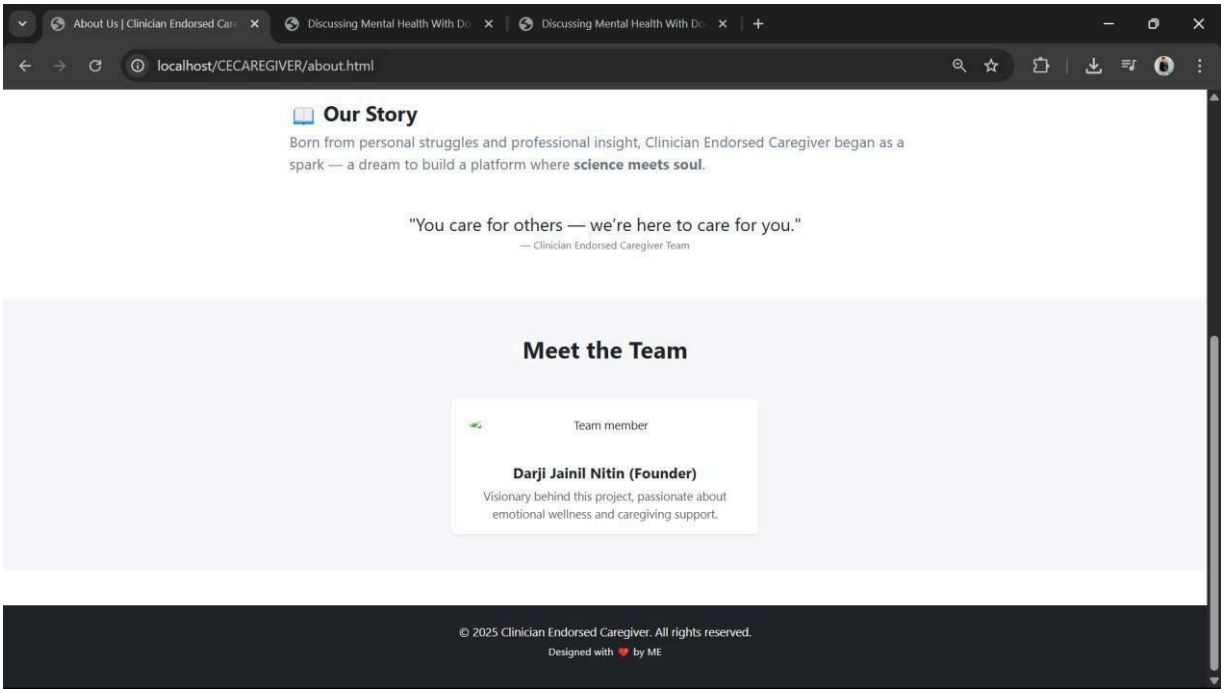


- **Donate Page :**

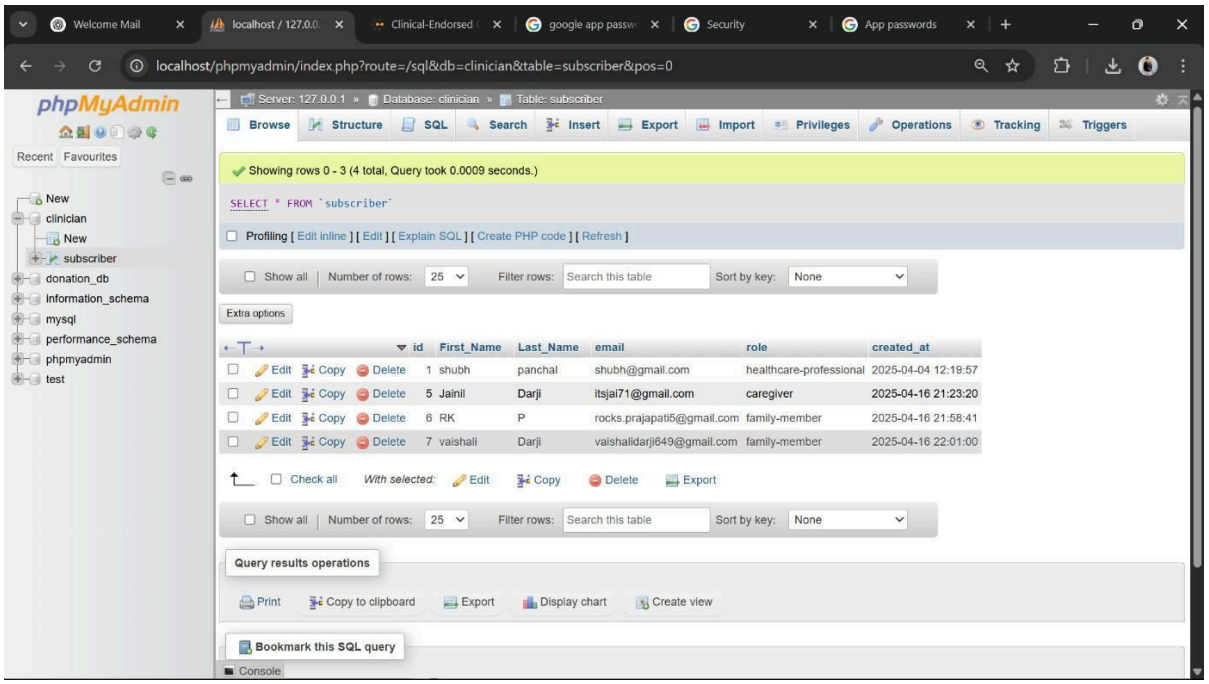


- **About Page :**

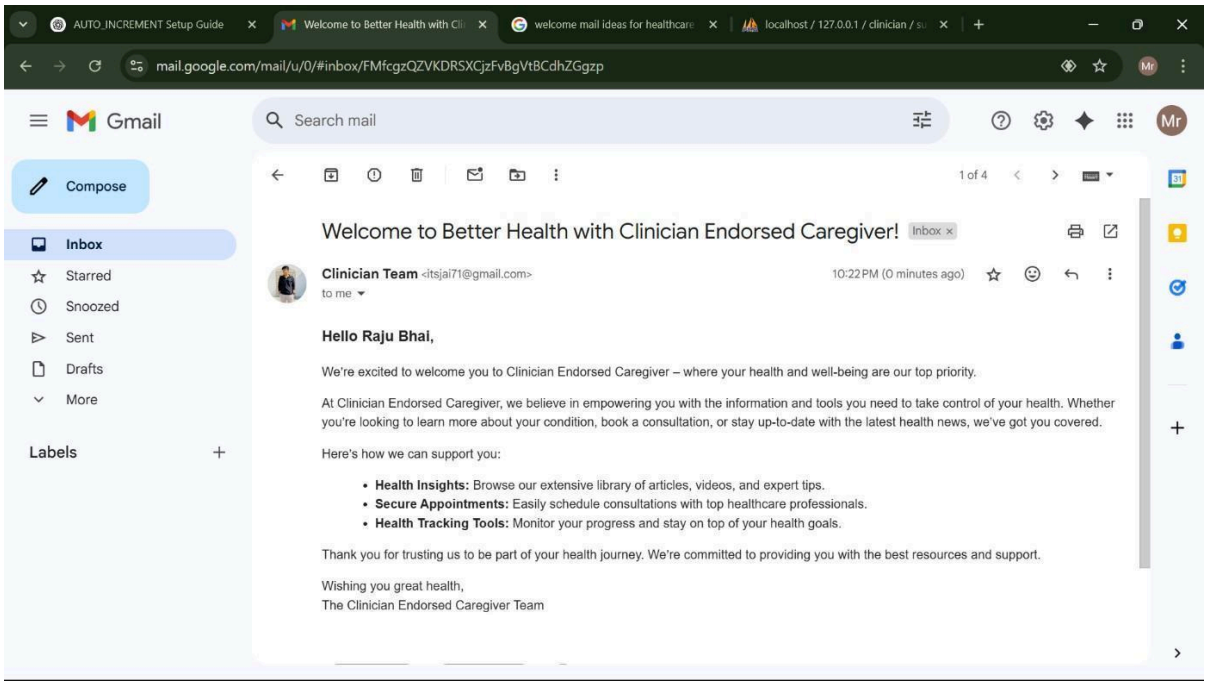




Database For Subscribers :



Welcome Mail :



CHAPTER: 10

ADVANTAGES

10.1 Advantages

- Efficiency: It provides a centralized platform where caregivers can easily access verified and clinician-approved information without searching multiple sources.

- Time Management: It saves time by offering structured and categorized content, helping users quickly find relevant caregiving resources.

- Resource Optimization: Ensures effective use of available information by organizing content in a clear and accessible manner.

- Real-world Use Case: Addresses a meaningful social problem by supporting caregivers with proper guidance and reliable information.

- User-Friendly Design: The website is designed with simple navigation, making it easy for users of all age groups to use.

- Full Stack Experience: The project uses frontend, backend, and database technologies, providing complete web development exposure.

- MySQL Integration: Helps in efficient storage and management of user data and caregiving resources.

- Bootstrap for Responsive Design: Ensures the website works smoothly on mobile, tablet, and desktop devices.

- Easy to Deploy on Most Servers: PHP and MySQL are widely supported, making deployment simple.

- Open-Source and Free Technologies: Uses free tools and technologies, making it cost-effective for development.

- Reduced Misinformation: Provides expert-reviewed content, reducing chances of incorrect caregiving information.

- Expandable Project: The system can be extended in future with features like admin panel, feedback system, and additional modules.

10.2 Limitations

- Dependency on Internet: The system requires internet access, limiting usability in offline conditions.
- Limited Advanced Features: Does not include features like live chat, AI recommendations, or real-time updates.
- Security Limitations: Basic security measures are implemented; advanced security is required for real-world use.
- Integration Limitations: Does not currently support integration with external healthcare systems or APIs.
- Scalability Constraints: Performance may be affected if the number of users and data increases significantly.
- Maintenance Challenges: Requires manual updates and maintenance of content and system.
- Limited User Interaction: Does not include interactive features like discussion forums or user communities.
- User Experience Constraints: Limited dynamic functionality due to minimal use of advanced scripting or real-time features.

CHAPTER: 11

CONCLUSION AND FUTURE ENHANCEMENT

11.1 Conclusion

This project aimed to design and develop a caregiving support website that provides users with essential resources, information, and a platform for better communication and awareness.
Using core web technologies like HTML, CSS (Bootstrap), PHP, and MySQL, the system offers a responsive, functional, and easy-to-navigate environment for caregivers and those seeking assistance.

Throughout the development process, attention was given to ensuring clarity, simplicity, and usability, especially for users who may not be tech-savvy. The project not only enhances digital accessibility to caregiving resources but also encourages community involvement and support sharing. Core functionalities such as user registration, secure data handling, and content display were implemented with performance and security in mind.

While the project has certain limitations, such as the absence of advanced backend frameworks and limited automation, it successfully meets its core objectives. The current version serves as a strong prototype that can be scaled and enhanced in the future by integrating features like real-time chat, notifications, admin dashboards, or API integrations.

In conclusion, this website demonstrates the potential of web-based solutions in the field of caregiving and highlights how even simple tools, when implemented with empathy and purpose, can make a positive impact on society.

11.2 Future Enhancement

While the current version of the website fulfills its basic objectives, several enhancements can be implemented to improve functionality, scalability, and user experience in the future:

- **User Role Management:** Introduce multiple user roles such as admin, caregiver, and visitor, each with specific permissions and dashboards.
- **Real-time Communication:** Implement live chat or messaging features to enable instant communication between caregivers and users.
- **Content Management System (CMS):** Develop a backend CMS to allow admins to manage website content (blogs, articles, events) without editing the code.
- **Advanced Security Features:** Add features like two-factor authentication (2FA), CAPTCHA, and encrypted password storage to enhance user security.
- **Mobile Application Integration:** Create a complementary mobile app to extend accessibility and reach a wider audience.
- **Search and Filter Functionality:** Provide intelligent search, filters, and tags to help users quickly find relevant content and services.
- **Analytics and Reporting:** Integrate analytics to monitor user behavior and generate reports for improving service quality.
- **Multilingual Support:** Offer content in multiple languages to cater to a diverse user base.
- **Online Appointment Scheduling:** Allow users to schedule appointments or request services directly from the website.
- **Donation or Volunteer Module:** Add a section where users can donate to support caregiving initiatives or register as volunteers.

BIBLIOGRAPHY

a. Course Outcome

Sr. No.	CO statement	Marks % weightage	Chapter No.
CO-1	Undertake problem identification, formulation and solution	20%	**CHAPTER 1** INTRODUCTION
CO-2	Design engineering solutions to complex problems utilising a systematic approach and team work	30%	**CHAPTER 2** SYSTEM REQUIREMENTS **CHAPTER 3** DAILY TASK
CO-3	Communicate with engineers and the community at large in written and oral forms	20%	**CHAPTER 6** SYSTEM DESIGN **CHAPTER 10** ADVANTAGES **CHAPTER 11** CONCLUSION AND FUTURE ENHANCEMENT
CO-4	Demonstrate the knowledge and understanding of engineering and management principle and apply it to assigned project	30%	**CHAPTER 4** FRONT END OF SYSTEM **CHAPTER 5** BACK END OF SYSTEM **CHAPTER 6** SYSTEM DESIGN **CHAPTER 7** DATA DICTIONARY **CHAPTER 8** TESTING

b. Books:

- **Fundamental of Software Engineering**
  - Author: Rajib Mall
- **The Joy of PHP Programming: A Beginner's Guide to Programming Interactive Web Applications with PHP and MySQL**
  - Author: Alan Forbes
- **Head First PHP & MySQL**
  - Author: Lynn Beighley & Michael Morrison
- **SQL Server Database**
  - Author: Mr.Gaurav Desani

C. Web Reference:

- **W3Schools. "CSS Tutorial." [Online].**
  - Available: https://www.w3schools.com/css/default.asp.. [Accessed: February 1, 2024].
- **Caregiver Action Network (CAN)**
  - The website provides a variety of resources for caregivers, including guides, support, and training. Though it's not specifically focused on dementia, it has valuable content for caregivers of people with dementia. Available: https://caregiveraction.org
- **Free Code Camp. "Learn to code — for free." [Online].**
  - Available: https://www.freecodecamp.org/ [Accessed: March 15, 2024].
- **Stack Overflow. "Where Developers Learn, Share, & Build Careers." [Online].**
  - Available: https://stackoverflow.com/ [Accessed: April 2, 2024]